Bachelor of Science in Electrical and Electronic Engineering

EEE 400 (January 2025): Thesis

Deep Sparse-to-Dense Automotive Radar Perception for Super-Resolved Object Detection and Classification

Submitted by

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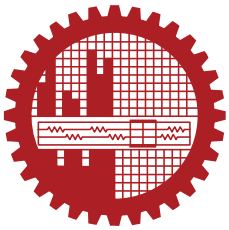
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June 2026

CANDIDATES' DECLARATION

This is to certify that the work presented in this thesis, titled, "Deep Sparse-to-Dense Automotive Radar Perception for Super-Resolved Object Detection and Classification", is the outcome of the investigation and research carried out by us under the supervision of Dr. Md. Asiful Islam.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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CERTIFICATION

This thesis titled, **"Deep Sparse-to-Dense Automotive Radar Perception for Super-Resolved Object Detection and Classification"**, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for EEE 400: Project/Thesis course, and as the requirements for the degree B.Sc. in Electrical and Electronic Engineering in June 2026.

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Thesis Outcome Evidence and Engineering Complexity Justification

Deep Sparse-to-Dense Automotive Radar Perception for Super-Resolved Object Detection and Classification

CO	Course Outcome Statement	Mapped PO	Evidence from Thesis Work	Thesis Reference	Attained?
CO1	Identify, formulate, research literature, interpret data, and analyze problems using principles of mathematical, natural and engineering sciences.	PO(b)	*The thesis identifies the limited angular resolution of sparse automotive radar as the central engineering problem and formulates it as a sparse-to-dense range–azimuth super-resolution problem followed by multi-view radar perception. The introduction defines the motivation, problem statement, research gap, and the distinction between Stage 1 super-resolution and Stage 2 perception. The results chapter interprets PSNR, SSIM, precision, recall, F1, mAP, and mAR to analyze the system performance.*	*Chapter 1, Sections 1.1–1.3; Chapter 2, Section 2.5; Chapter 4, Sections 4.1–4.4; Figure 3.1; Tables 4.1, 4.4, and 4.5.*	☐
CO2	Investigate of problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	PO(d)	*The thesis investigates the problem through a structured experimental methodology. Stage 1 uses Dataset3000, a paired sparse/dense simulated radar dataset, to train and evaluate an Attention U-Net for angular super-resolution. Stage 2 evaluates radar-based detection and classification using sequence-level data splitting, multi-view RA/RV/VA inputs, heatmap-based detection metrics, and qualitative prediction overlays. Conclusions are drawn from quantitative metrics and qualitative visualization while avoiding unsupported causal claims about SR causing all Stage 2 improvements.*	*Chapter 3, Sections 3.2–3.6; Chapter 4, Sections 4.1–4.4; Chapter 5, Sections 5.1–5.3; Figures 3.6, 4.2, 4.5, and 4.9.*	☐
CO3	Apply knowledge of mathematics, science, and engineering to solve research problems	PO(a)	*The thesis applies FMCW radar theory, MIMO virtual-array processing, FFT-based range/Doppler/angle processing, radar signal synthesis, deep neural network modelling, loss-function design, and evaluation metrics. Mathematical formulations include beat frequency, Doppler frequency, spatial phase, sparse-to-dense mapping, super-resolution loss, focal loss, and detection matching criteria.*	*Chapter 3, Sections 3.2–3.5; Equations 3.4–3.6, 3.2, 3.3, 3.11, and 4.6; Tables 3.1, 3.2, and 3.6.*	☐
CO4	Demonstrate adherence to professional ethics and ethical principles of research	PO(h)	*The thesis maintains responsible research practice by citing prior work, clearly separating original contributions from adapted methods, using controlled datasets and documented processing pipelines, and honestly reporting limitations. The relationship between super-resolution and downstream detection is framed cautiously, and the thesis avoids claiming that SR alone caused the Stage 2 detection performance.*	*Chapter 2, Sections 2.1–2.5; Chapter 4, Section 4.4; Chapter 5, Sections 5.1–5.2;Bibliography; iThenticate report:* ***[Insert report reference or appendix location]****.*	☐

Continued on next page

CO	Course Outcome Statement	Mapped PO	Evidence from Thesis Work	Thesis Reference	Attained?
CO5	Communicate effectively with oral presentation and detailed thesis report	PO(j)	*The thesis communicates the research through a structured report containing introduction, literature review, methodology, results, limitations, future work, and conclusion. Figures, tables, equations, and result visualizations are used to explain the two-stage pipeline, Dataset3000 generation, Attention U-Net, RAMP-CNN+, FullModel, training curves, and detection/classification results.*	*Full thesis report; Chapter 3 figures and tables; Chapter 4 result figures and tables; Presentation slides:* ***[Insert slide numbers after final slide preparation]****.*	☐
CO6	Research contemporary aspects related to the problem and topic through detailed literature review and understand	PO(l)	*The literature review covers automotive FMCW/MIMO radar, angular-resolution limitations, sparse-array super-resolution, classical methods such as compressed sensing, MUSIC and ESPRIT, deep-learning-based radar super-resolution, RAMP-CNN/RODNet-style radar detection, multi-view fusion, and radar road-user classification using micro-Doppler and time-frequency features.*	*Chapter 2, Sections 2.1–2.5; Bibliography; Chapter 5, Section 5.2.*	☐

Knowledge Profile, Complex Engineering Problem, and Complex Engineering Activity Justification

Code	Meaning	Thesis-Specific Justification
Knowledge Profile Attributes		
K1	Knowledge of natural sciences applicable to the engineering discipline.	*The thesis uses electromagnetic wave propagation concepts underlying FMCW radar sensing, including reflected chirps, propagation delay, range-dependent path loss, Doppler shift, and angular phase variation across antenna elements. These principles support the radar signal model and the interpretation of range, velocity, and azimuth information.*
K2	Conceptually based mathematics, numerical analysis, statistics, and formal aspects of computer and information science supporting analysis and modelling.	*The work uses FFT-based signal processing, sparse-to-dense mapping, supervised image-regression, PSNR, SSIM, precision, recall, F1, mAP, mAR, focal loss, cross-entropy loss, supervised contrastive learning, and neural-network optimization. Numerical simulation and deep-learning training are central to both Stage 1 and Stage 2.*
K3	Engineering fundamentals required for the engineering discipline.	*The thesis applies electrical and electronic engineering fundamentals including FMCW radar operation, MIMO virtual-array formation, range/Doppler/angle processing, ADC cube representation, and radar signal synthesis. These concepts are used to generate RA, RV, and VA radar views and to design the radar perception pipeline.*
K4	Specialist knowledge that provides theoretical frameworks and bodies of knowledge for accepted practice areas in the engineering discipline.	*Specialist knowledge is used in sparse-array radar imaging, radar angular super-resolution, deep-learning-based radar perception, multi-view radar fusion, heatmap-based detection, prototype-based classification, and radar-camera feature alignment. The methodology is informed by radar super-resolution literature and RAMP-CNN/RODNet-style radar detection literature.*

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Code	Meaning	Thesis-Specific Justification
K5	Engineering design knowledge that supports the design of complex engineering solutions.	*The thesis designs and implements a two-stage radar perception pipeline. Stage 1 uses an Attention U-Net for sparse-to-dense angular super-resolution. Stage 2 uses RAMP-CNN+ for heatmap-based detection and* `FullModel` *for prototype-based classification and object counting. The design balances angular resolution, cost, memory, and detection/classification performance.*
K6	Knowledge of engineering practice, including technology, tools, resources, safety, standards, and practical constraints.	*The work uses CARLA, MATLAB FMCW signal synthesis, Python/PyTorch notebooks, Kaggle GPU resources, AWR1843 radar data, cached radar-view generation, mixed-precision training, checkpointing, and practical GPU/memory/runtime constraints.*
K7	Comprehension of the role of engineering in society, including ethics, sustainability, economy, environment, health, safety, and public welfare.	*The thesis is motivated by low-cost autonomous-driving perception and road-user safety. Radar perception can support safer detection of pedestrians, cars, and cyclists under challenging lighting and weather conditions. The work also considers cost–resolution tradeoffs relevant to affordable automotive sensing.*
K8	Engagement with research literature in the discipline.	*The thesis reviews and uses literature on automotive radar signal processing, MIMO radar angular estimation, compressed sensing, MUSIC, ESPRIT, radar super-resolution, RAMP-CNN, RODNet, radar-camera fusion, and radar road-user classification. The literature review informs the research gap and methodology.*
Complex Engineering Problem Attributes		
P1	Depth of knowledge required: the problem cannot be solved without applying principles from mathematics, natural sciences, engineering fundamentals, and specialist engineering knowledge.	*The problem requires knowledge of radar physics, FMCW signal processing, MIMO virtual arrays, angular resolution, deep neural networks, image-regression metrics, detection metrics, and multi-view radar perception. It is therefore not a routine application of a single textbook method.*
P2	Range of conflicting requirements: the problem involves wide-ranging or conflicting technical, performance, cost, safety, environmental, social, or practical requirements.	*The thesis addresses the tradeoff between radar hardware cost and angular resolution. A dense array improves angular separation but increases hardware complexity and cost, while a sparse array is cheaper but less accurate. The methodology also balances model performance, GPU memory, runtime, and practical feasibility.*
P3	Depth of analysis required: the problem needs analysis, modelling, simulation, experiment, interpretation, or judgment beyond a standard textbook solution.	*The work includes simulation-based Dataset3000 generation, FMCW radar modelling, sparse/dense radar-pair synthesis, Attention U-Net training, RAMP-CNN+ detection, prototype-based classification, quantitative metric analysis, qualitative radar visualization, and interpretation of limitations.*
P4	Familiarity of issues: the problem may include infrequently encountered, partially defined, emerging, or research-oriented issues.	*Sparse-to-dense radar super-resolution combined with downstream radar perception is an emerging research-oriented problem. The thesis also handles sim-to-real differences, geometry-specific SR training, heatmap-based radar detection, and prototype-based radar classification.*
P5	Extent of applicable codes: the problem may require codes, standards, engineering practices, or literature that are not fully sufficient by themselves.	*No single standard or code fully defines the proposed radar SR and perception pipeline. The thesis relies on research literature, dataset design, radar-processing practice, and experimental evaluation rather than applying a fixed established design rule.*
P6	Extent of stakeholder involvement and conflicting requirements: the problem may involve users, supervisors, clients, society, industry, or other stakeholders with different needs.	*The work is relevant to road users, autonomous-vehicle developers, and safety-focused transportation systems. It also considers supervisor feedback, thesis requirements, computational limits, low-cost sensing needs, and safe perception requirements.*

Continued on next page

Code	Meaning	Thesis-Specific Justification
P7	Interdependence: the problem has multiple connected components, variables, sub-systems, or constraints that influence one another.	*The system contains connected stages: radar ADC processing, RA/RV/VA generation, Stage 1 SR, Stage 2 detection/classification, heatmap generation, loss design, dataset splitting, and evaluation. Changes in radar representation, class balance, thresholding, or model architecture affect final perception performance.*
Complex Engineering Activity Attributes		
A1	Range of resources: the activity uses diverse resources such as people, equipment, software, data, literature, laboratory facilities, or engineering tools.	*The thesis uses multiple resources, including research literature, CARLA simulation, MATLAB FMCW radar signal synthesis, AWR1843 radar data, synchronized camera images, Python/PyTorch deep-learning implementation, Kaggle GPU notebooks, LaTeX report writing, and supervisor feedback.*
A2	Level of interaction: the activity requires interaction among tasks, tools, systems, people, or disciplines.	*The work integrates radar signal processing, simulation, deep learning, dataset preparation, model training, validation, visualization, and thesis documentation. It also connects Stage 1 super-resolution with Stage 2 radar perception while keeping the evaluation of the two stages distinct.*
A3	Innovation: the activity involves creative use of engineering knowledge, adaptation of methods, new design choices, or non-routine problem solving.	*The thesis adapts sparse-array radar super-resolution to a specific low-cost automotive radar configuration, develops Dataset3000 for paired sparse/dense radar training, implements Attention U-Net for SR, extends RAMP-CNN-style perception through RAMP-CNN+, and adds a prototype-based classification and counting pathway.*
A4	Consequences for society and environment: the activity may have effects on health, safety, sustainability, environment, economy, or society.	*The work is relevant to safer autonomous-driving perception, particularly for detecting pedestrians, cars, and cyclists. The sparse-to-dense approach also addresses the economic need for low-cost radar hardware while attempting to improve perception capability.*
A5	Familiarity: the activity may require work in unfamiliar, uncertain, or changing contexts.	*The thesis involves research-oriented and unfamiliar tasks including custom radar dataset synthesis, sim-to-real qualitative validation, deep radar super-resolution, multi-view radar heatmap detection, prototype-based radar classification, and interpretation under limited dataset and compute constraints.*

I/We, the undersigned, pledge that I/we filled out the tables with consultation and feedback of my/our supervisor.

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The relevant tables are verified based on the thesis report, presentation, demonstration, viva voce, and assessment evidence. Any suggested modifications are reflected in the table.

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LIST OF ALGORITHMS

ABSTRACT

Automotive radar is an important sensing modality for autonomous driving because it provides range, velocity, and angular information under challenging lighting and weather conditions. However, low-cost sparse radar arrays suffer from limited angular resolution, which makes it difficult to separate nearby road users in the range–azimuth domain. Reliable radar perception therefore requires both improved angular representation and effective multi-view detection and classification of safety-relevant objects such as pedestrians, cars, and cyclists.

This thesis presents a two-stage radar perception pipeline for low-cost autonomous-driving applications. In Stage 1, an Attention U-Net is used for radar angular super-resolution, mapping sparse-array range–azimuth features from a $2\mathrm{Tx} \times 4\mathrm{Rx}$ configuration to a dense-array-equivalent $5\mathrm{Tx} \times 10\mathrm{Rx}$ representation. To train this model, Dataset3000 is developed as a custom paired simulated radar dataset using CARLA scenes and a MATLAB FMCW signal-synthesis model, producing registered sparse and dense radar pairs. In Stage 2, road-user perception is performed using multi-view radar representations. A heatmap-based RAMP-CNN+ model is used for detection and classification, while a prototype-based `FullModel` provides an additional classification and object-counting pathway.

The Stage 1 super-resolution model achieved a test PSNR of 40.10 dB and SSIM of 0.8716, and qualitative evaluation on measured radar data showed cleaner and more focused angular peaks compared with the raw sparse-array output. On a three-class AWR1843 radar dataset containing person, car, and cyclist classes, the Stage 2 RAMP-CNN+ detector achieved a weighted F1 score of 0.717, mAP of 0.722, and mAR of 0.682 using the SR-enhanced RA input, compared with weighted F1 of 0.468, mAP of 0.480, and mAR of 0.424 using the raw-RA input. Overall, the proposed pipeline demonstrates the feasibility of combining learned angular enhancement with multi-view radar perception for low-cost autonomous-driving systems.

MAIN BODY

CHAPTER 1: INTRODUCTION

Autonomous driving systems require reliable perception of the surrounding environment. A vehicle must be able to detect, localize, and classify nearby road users such as pedestrians, vehicles, and cyclists before safe planning and control decisions can be made. This thesis investigates a two-stage radar-based perception pipeline in which the first stage performs radar angular super-resolution and the second stage performs road-user detection, classification, and object counting.

1.1 Background and Motivation

Modern autonomous-driving perception systems commonly rely on multiple sensing modalities, including cameras, lidar, and radar. Cameras provide rich visual and semantic information, but their performance can degrade under poor illumination, glare, fog, rain, or other adverse visibility conditions. Lidar provides accurate geometric measurements, but it is also affected by cost, environmental scattering, and deployment constraints. Radar, particularly millimetre-wave FMCW radar, offers an important complementary sensing modality because it can directly measure range and radial velocity and remains useful under challenging illumination and weather conditions where camera- or lidar-based perception may degrade [1, 2].

Despite these advantages, automotive radar has its own limitations. One of the most important limitations is angular resolution. In a MIMO radar system, angular resolution is strongly related to the effective virtual aperture, which depends on the number and arrangement of transmit and receive antenna elements. A dense antenna array can produce sharper angular separation, but it requires more antenna channels, increasing hardware cost and system complexity. This creates a practical engineering tradeoff between sensing performance and low-cost automotive deployment.

This thesis addresses this tradeoff by proposing a learned super-resolution front-end for radar range–azimuth data. Instead of relying only on a physically dense radar array, the first stage learns to reconstruct a dense-array-equivalent range–azimuth representation from a sparse low-cost radar configuration. The studied sparse configuration is a $2\mathrm{Tx} \times 4\mathrm{Rx}$ array with 8 virtual elements, while the dense target configuration is a $5\mathrm{Tx} \times 10\mathrm{Rx}$ array with 50 virtual elements. The goal is to improve effective angular detail while retaining a hardware configuration closer to low-cost automotive radar.

The second stage of the thesis investigates whether multi-view radar representations can be used for road-user perception. Range–azimuth (RA), range–velocity (RV), and velocity–azimuth

(VA) views provide complementary spatial and motion information. By combining these radar views with deep learning models, the system aims to detect and classify person, car, and cyclist classes from radar data. The overall motivation is therefore twofold: to improve angular representation from sparse radar data, and to use multi-view radar information for downstream road-user perception.

1.2 Problem Statement

Low-cost automotive radar systems are attractive for practical deployment, but their sparse antenna configurations limit angular resolution. This limitation can make it difficult to separate nearby objects in the range–azimuth domain and can introduce side-lobe or aliasing artifacts. Increasing the number of antenna channels can improve angular resolution, but it also increases hardware cost and complexity.

The first problem addressed in this thesis is whether a deep neural network can recover dense-array-equivalent angular information from sparse-array radar measurements. More specifically, the work studies whether a super-resolution model can learn a mapping from the RA representation of a $2\mathrm{Tx} \times 4\mathrm{Rx}$ sparse MIMO array to the RA representation of a $5\mathrm{Tx} \times 10\mathrm{Rx}$ dense MIMO array. This problem is formulated as a supervised image-regression task in the range–azimuth domain.

The second problem addressed in this thesis is radar-based road-user perception from multi-view radar data. Given RA, RV, and VA representations derived from FMCW radar, the task is to detect, classify, and count road users belonging to three retained classes: person, car, and cyclist. This thesis investigates two Stage 2 model designs for this purpose. The first is a heatmap-based detection model, RAMP-CNN+, which localizes objects using class-wise radar heatmaps. The second is a prototype-based classification and counting model, `FullModel`, which produces scene-level class logits and object-count predictions.

The overall research question can therefore be stated as follows: can a two-stage radar pipeline combine learned angular super-resolution with multi-view radar perception to support low-cost autonomous-driving sensing? In answering this question, the thesis keeps the evidence for the two stages distinct. Stage 1 is evaluated as an angular super-resolution module using image-quality metrics and qualitative real-world comparison. Stage 2 is evaluated as a radar perception system using detection, classification, and counting metrics. The RAMP-CNN+ detector is evaluated using both raw RA input and SR-enhanced RA input under the same evaluation protocol. However, the thesis does not make an unsupported general claim that super-resolution alone determines downstream detection performance beyond the implemented dataset, model,

and array-geometry setting.

1.3 Contributions

The main contributions of this thesis are summarized below.

1. **Dataset3000: a paired simulated radar dataset for super-resolution.** This thesis develops Dataset3000, a custom paired simulated radar dataset generated using CARLA scenes and a bespoke MATLAB FMCW signal-synthesis model. For each frame, the same target list is rendered through both a sparse $2\mathrm{Tx} \times 4\mathrm{Rx}$ radar configuration and a dense $5\mathrm{Tx} \times 10\mathrm{Rx}$ radar configuration. This produces perfectly registered sparse/dense ADC pairs for supervised radar angular super-resolution. The dataset spans multiple CARLA maps, ego spawn points, and weather conditions, and stores the low-configuration radar cube, high-configuration radar cube, synchronized camera image, and target labels.

2. **Stage 1 radar angular super-resolution using an Attention U-Net.** A deep super-resolution front-end is implemented to reconstruct dense-array-equivalent range–azimuth maps from sparse-array radar inputs. The model uses a five-channel input representation containing beamsteering magnitude and phase together with spatial-covariance features. The Stage 1 model achieves a test PSNR of 40.10 dB and test SSIM of 0.8716, and qualitatively improves measured real-world radar frames by producing cleaner and more focused angular peaks.

3. **Stage 2 heatmap-based radar detection using RAMP-CNN+.** This thesis implements RAMP-CNN+, a multi-view radar detection and classification model for person, car, and cyclist classes. The model uses RA, RV, and VA radar representations and includes complex-valued RA processing, cross-modal feature alignment, squeeze-and-excitation adaptive fusion, and a class-balanced asymmetric focal loss. With the SR-enhanced RA input, the heatmap-based detector achieves weighted F1 of 0.717, mAP of 0.722, and mAR of 0.682 on the held-out test set. In comparison, the raw-RA configuration achieves weighted F1 of 0.468, mAP of 0.480, and mAR of 0.424. Thus, within the implemented experimental setting, the SR-enhanced configuration shows improved detection performance.

4. **Stage 2 prototype-based classification and object-counting model.** A second Stage 2 model, referred to as `FullModel`, is developed for scene-level radar classification and object counting. This model combines a `MultiScaleObjectCounter` branch with a projection-based `PrototypeClassifier` branch. It encodes multiple radar projections, applies cross-projection attention, and uses exponential-moving-average class prototypes to stabilize classification under class imbalance.

5. **Reproducible evaluation with leakage-free sequence-level splitting.** The Stage 2 experiments use a manually curated $65/20/15$ sequence-level split so that frames from the same driving sequence do not appear in multiple splits. This reduces temporal leakage and provides a more defensible evaluation protocol. The thesis reports both quantitative metrics and qualitative visualizations, including Stage 1 super-resolution examples, raw-RA versus SR-enhanced RAMP-CNN+ comparison, Stage 2 training curves, per-class detection results, and prediction overlays.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant work on MIMO radar angular resolution, radar super-resolution, and radar-based object detection and classification. Chapter 3 describes the proposed two-stage methodology, including FMCW radar signal processing, Stage 1 super-resolution, Stage 2 detection and classification, the prototype-based classification model, dataset preparation, and design considerations. Chapter 4 presents the experimental setup, quantitative results, qualitative visualizations, and discussion of the Stage 1 and Stage 2 results. Finally, Chapter 5 discusses limitations, future work, and the concluding findings of the thesis.

CHAPTER 2: LITERATURE REVIEW

This chapter reviews the research areas most relevant to the proposed two-stage radar perception pipeline. The discussion is organized into five themes: MIMO radar and angular resolution, radar super-resolution, radar object detection, radar classification of road users, and the research gap addressed in this thesis. The purpose of this chapter is to position the present work within existing radar signal-processing and deep-learning-based automotive radar literature.

2.1 MIMO Radar and Angular Resolution

Millimetre-wave frequency-modulated continuous-wave (FMCW) radar is widely used in automotive sensing because it can estimate range and radial velocity and can operate under conditions where purely optical sensors may degrade. In FMCW radar, target range is obtained from the beat frequency generated by mixing the transmitted and received chirps, while radial velocity is obtained from Doppler phase variation across repeated chirps. Automotive radar signal processing typically includes range processing, Doppler processing, angle estimation, detection, and tracking stages [1].

Radar is also valuable in adverse environmental conditions. Camera-based systems may be affected by illumination changes, glare, darkness, fog, rain, or snow, while radar can continue to provide range and velocity measurements in many such conditions. Fusion of radar and camera information has therefore been studied for robust target recognition and tracking in self-driving applications, especially under severe weather conditions [2]. This makes radar an important sensing modality for autonomous-driving perception.

Multiple-input multiple-output (MIMO) radar improves angular sensing by using multiple transmit and receive antennas to form a larger virtual array. If a radar system has N_{Tx} transmit antennas and N_{Rx} receive antennas, the virtual array can contain up to $N_{\mathrm{Tx}}N_{\mathrm{Rx}}$ virtual elements, depending on the array geometry and waveform design. The phase differences across these virtual antenna elements are used to estimate target direction. Direction-of-arrival (DOA) and direction-of-departure (DOD) estimation methods for MIMO radar have been widely studied, including reduced-dimension MUSIC and improved joint ESPRIT approaches [3, 4].

Angular resolution is closely related to the effective aperture of the antenna array. A larger virtual aperture can form a narrower main lobe and can better separate targets located at similar ranges but different azimuth angles. However, increasing the number of transmit and receive channels increases hardware cost, system complexity, physical layout constraints, and calibration requirements. This creates a practical engineering tradeoff in automotive radar: high an-

gular resolution is desirable for perception, but dense antenna hardware is more expensive and less attractive for low-cost deployment.

Sparse arrays reduce hardware complexity but lead to coarser angular resolution and stronger side-lobe or aliasing effects. These limitations are especially important in the range–azimuth domain, where poor angular resolution can make it difficult to separate nearby road users. The central motivation of this thesis follows directly from this tradeoff. The sparse radar configuration used as the low-cost input is a $2\mathrm{Tx} \times 4\mathrm{Rx}$ MIMO array with 8 virtual elements, while the dense target configuration for super-resolution is a $5\mathrm{Tx} \times 10\mathrm{Rx}$ MIMO array with 50 virtual elements. The first stage of this thesis investigates whether a learned model can reconstruct dense-array-equivalent angular structure from the sparse configuration.

2.2 Radar Super-Resolution

Radar super-resolution refers to methods that attempt to recover finer spatial or angular detail than that provided directly by conventional FFT-based processing. In the angular domain, the objective is to sharpen directional responses and separate targets that are difficult to distinguish using a limited aperture. This problem is directly relevant to low-cost automotive radar because sparse arrays provide fewer virtual antenna elements and therefore lower angular resolution.

Classical radar super-resolution and direction-estimation methods include compressed sensing, MUSIC, and ESPRIT. Compressed sensing exploits sparsity in the target scene and formulates radar imaging as a sparse recovery problem. Potter et al. discuss the use of sparsity and compressed sensing in radar imaging, showing how sparse reconstruction can be used to recover radar scenes from limited measurements [5]. Experimental evaluation of compressive sensing for direction-of-arrival estimation in automotive radar has also been reported [6]. MUSIC and ESPRIT are subspace-based methods that estimate angular components from the structure of the received signal covariance matrix. Reduced-dimension MUSIC and improved joint ESPRIT methods have been applied to MIMO radar DOD/DOA estimation problems [3, 4].

Although these classical methods are theoretically powerful, their performance can depend strongly on signal-to-noise ratio, scene sparsity, calibration accuracy, number of targets, and validity of the assumed signal model. Practical automotive radar scenes may contain extended targets, multipath reflections, near-field effects, non-ideal antenna patterns, and clutter. These factors can make purely model-based super-resolution difficult to apply robustly in complex driving environments.

Deep-learning-based radar super-resolution provides an alternative approach by learning a mapping from low-resolution radar measurements to high-resolution target representations. Instead of explicitly solving a sparse inverse problem or estimating signal subspaces, a neural network is trained using paired data to reconstruct sharper radar images. This approach is attractive when

paired sparse and dense radar data can be generated through simulation, because the model can learn to suppress side-lobe artifacts and recover dense-array-like angular responses.

The Stage 1 method in this thesis is based on the work of Schuessler et al., who proposed super-resolution radar imaging with sparse arrays using a deep neural network trained with enhanced virtual data [7]. Their work provides the core methodological basis for learning sparse-array radar image enhancement using deep learning. The present thesis adapts this direction to a specific automotive radar configuration, mapping a sparse $2\mathrm{Tx} \times 4\mathrm{Rx}$ input representation to a dense $5\mathrm{Tx} \times 10\mathrm{Rx}$ target representation.

A key challenge in supervised radar super-resolution is obtaining paired low- and high-resolution ground truth. Measured dense-array ground truth is difficult to collect in perfect registration with sparse-array measurements. This thesis addresses that issue by constructing Dataset3000, a custom paired simulated radar dataset in which both sparse and dense radar configurations are synthesized from the same CARLA target list. This provides the paired supervision needed to train the Stage 1 Attention U-Net while still allowing qualitative sim-to-real validation on independently measured radar data.

2.3 Radar Object Detection

Radar object detection aims to localize and classify road users from radar measurements. Compared with camera images, radar representations are generally less dense and less semantically rich. However, radar provides direct range and velocity information, making it valuable for detecting moving objects in autonomous-driving scenes. Deep learning has therefore been increasingly applied to radar representations such as range–azimuth, range–Doppler, range–velocity, and velocity–azimuth maps.

Convolutional neural networks can learn features directly from radar tensors. A range–azimuth representation provides spatial information in the range and angular directions, while a range–Doppler or range–velocity representation provides motion information. Since no single radar view contains all relevant information, multi-view radar perception has become an important design direction. RAMP-CNN introduced a neural network architecture for enhanced automotive radar object recognition using radar multi-perspective information [8]. This supports the idea that combining different radar views can improve object recognition compared with relying on a single view.

RODNet is another important radar object detection framework. Wang et al. proposed a real-time radar object detection network cross-supervised by camera-radar fused object 3D localization [9]. Their work demonstrated that radar object detection can be formulated using neural networks that predict object confidence maps from radar data, and that cross-modal supervision can improve radar feature learning. This is particularly relevant to the Stage 2 RAMP-CNN+

model in this thesis, which also uses heatmap-based radar detection and camera-guided feature alignment during training.

Radar-camera fusion has also been studied for multi-view 3D object detection. Wu et al. proposed MVFusion, a multi-view 3D object detection method using semantic-aligned radar and camera fusion [10]. While the specific model in this thesis is different, the broader idea of using complementary radar and camera information supports the cross-modal design of the Stage 2 model.

The RAMP-CNN+ detector in this thesis follows the heatmap-based and multi-view radar perception direction. It uses RA, RV, and VA radar views, processes the complex RA input with a complex-valued convolutional branch, aligns radar and camera embeddings through cross-modal feature alignment, fuses radar views using adaptive attention, and trains the heatmap head using a class-balanced asymmetric focal loss. The model is designed for three-class road-user detection and classification: person, car, and cyclist.

2.4 Radar Classification of Road Users

Radar classification of road users focuses on distinguishing object categories from their radar signatures. This task is challenging because radar does not directly capture visual appearance such as colour or texture. Instead, classification must rely on range, velocity, angle, amplitude, temporal variation, and radar cross-section-related information. Road users such as pedestrians, vehicles, and cyclists can produce different radar signatures because of differences in size, motion pattern, and scattering behaviour.

Micro-Doppler signatures are widely used for radar-based classification. Moving parts of a target, such as human limbs or bicycle wheels, create additional Doppler components around the main target velocity. These components can provide discriminative information for separating pedestrians, bicyclists, and vehicles. Du et al. studied pedestrian and bicyclist identification using micro-Doppler signatures under different approaching aspect angles [11]. Their work supports the idea that velocity-domain radar signatures contain class-relevant information.

Classical radar classification methods often use hand-crafted features followed by machine-learning classifiers. These features may be extracted from time-frequency representations, Doppler spread, target extent, or other radar-domain statistics. Molchanov et al. used joint time-frequency analysis for classification of ground moving radar targets [12]. Such approaches show that radar target classification can benefit from temporal and spectral structure, even without image-like semantic information.

Moment-space and distribution-based classification methods are also relevant to radar-based road-user recognition. Instead of relying only on a single peak or detection cell, these methods represent the distribution of radar energy across range, velocity, or angle. Moments and dis-

tribution statistics can describe spread, asymmetry, and temporal variation of radar responses. This is useful because real road users are not ideal point targets; they produce extended and distributed radar returns.

The prototype-based `FullModel` developed in this thesis is motivated by these radar classification ideas. It encodes six projection views: `ra_range`, `ra_azimuth`, `rv_range`, `rv_velocity`, `va_azimuth`, and `va_velocity`. These projections reflect the physical intuition that range, azimuth, and velocity carry different class-discriminative information. Velocity projections can capture motion spread and micro-Doppler-like behaviour, while azimuth projections can describe angular extent and spatial spread. The model combines these projection features using cross-projection attention and stabilizes classification through exponential-moving-average class prototypes.

2.5 Research Gap

The reviewed literature shows that MIMO radar angular resolution, radar super-resolution, radar object detection, and radar road-user classification have all been studied as important research directions. Automotive radar signal-processing literature explains the importance of FMCW processing, virtual arrays, and angle estimation [1]. Classical direction-estimation and super-resolution methods such as compressed sensing, MUSIC, and ESPRIT provide mathematical tools for high-resolution radar imaging and DOA estimation [3–6]. Deep-learning-based radar super-resolution extends this idea by learning sparse-to-dense radar image mappings from paired data [7]. Radar detection models such as RAMP-CNN and RODNet show that neural networks can learn object-recognition and object-detection features from radar views [8, 9]. Radar classification studies further show that micro-Doppler and time-frequency signatures carry useful information for distinguishing road users [11, 12].

However, these research directions are often treated separately. Based on the literature reviewed in this thesis, there remains a gap in combining a geometry-specific sparse-to-dense radar super-resolution front-end with downstream multi-view radar detection and classification on a low-cost automotive radar hardware class. In particular, the configuration studied in this thesis maps a sparse $2\mathrm{Tx} \times 4\mathrm{Rx}$ radar representation toward a dense $5\mathrm{Tx} \times 10\mathrm{Rx}$ target representation and then uses multi-view radar representations for road-user perception.

This thesis addresses that gap through a two-stage radar perception pipeline. Stage 1 performs angular super-resolution in the range–azimuth domain using an Attention U-Net trained on paired sparse/dense radar data. Stage 2 performs road-user perception using multi-view radar representations through two models: RAMP-CNN+ for heatmap-based detection and `FullModel` for prototype-based classification and object counting. The thesis also constructs Dataset3000 for supervised radar super-resolution and uses a sequence-level evaluation protocol

for the Stage 2 dataset.

At the same time, the thesis frames the relationship between the two stages carefully. Stage 1 is evaluated through super-resolution metrics and qualitative measured-data examples, while Stage 2 is evaluated through radar perception metrics. The implemented RAMP-CNN+ detector is evaluated using both raw RA input and SR-enhanced RA input under the same evaluation protocol. However, this comparison is still limited to the specific dataset, model, and array-geometry setting used in this thesis. Broader raw-versus-SR evaluation across additional scenes, radar geometries, and repeated training runs remains a future research direction. This framing preserves the contribution of each stage without overstating causal claims beyond the current experimental scope.

CHAPTER 3: METHODOLOGY

3.1 System Overview

The proposed methodology consists of a two-stage radar perception pipeline for autonomous-driving applications. The first stage performs radar angular super-resolution in the range–azimuth domain, while the second stage performs road-user detection, classification, and object counting using multi-view radar representations.

The input to the pipeline is a raw FMCW MIMO radar ADC cube. Standard radar signal-processing operations are first applied to generate range–azimuth (RA), range–velocity (RV), and velocity–azimuth (VA) representations. For the SR-enhanced pipeline, the RA representation is passed through the Stage 1 super-resolution model, which enhances angular resolution using an Attention U-Net. The resulting SR-enhanced RA map is combined with the RV and VA representations and passed to the Stage 2 perception models. For comparison, a raw-RA pipeline is also evaluated, where the Stage 2 detector receives the standard FFT-based RA representation without the super-resolution module. The heatmap-based RAMP-CNN+ model performs object detection and classification, while the prototype-based `FullModel` provides an additional classification and object-counting pathway.

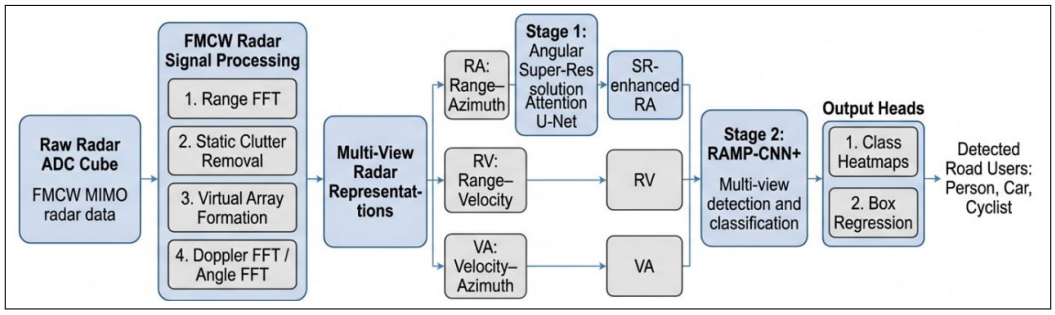


Figure 3.1: Overall system architecture of the proposed two-stage radar perception pipeline.

3.2 FMCW Radar Signal Processing

The radar front-end used in this thesis is based on frequency-modulated continuous-wave (FMCW) multiple-input multiple-output (MIMO) radar. In an FMCW radar, each transmitted chirp is reflected by objects in the scene and received after a propagation delay. The received signal is mixed with the transmitted chirp to produce an intermediate-frequency beat signal. The beat frequency is related to the target range, while the phase variation across repeated chirps is related to radial velocity. In a MIMO radar, the signals received across the transmit–receive antenna

combinations are also used to form a virtual antenna array, from which angular information can be estimated.

The raw radar measurement for each frame is represented as a complex ADC cube. For the AWR1843 radar data used in the detection stage, each frame has the shape $128 \times 255 \times 4 \times 2$, corresponding to fast-time ADC samples, slow-time chirps, receive antennas, and transmit antennas, respectively. With time-division multiplexed MIMO operation, the 4 receive antennas and 2 transmit antennas form an 8-element virtual array. This sparse virtual aperture is the hardware setting for which the angular resolution is limited, motivating the super-resolution stage described in Section 3.3.

The radar processing chain converts each raw ADC cube into three two-dimensional radar views: range–azimuth (RA), range–velocity (RV), and velocity–azimuth (VA). First, a range fast Fourier transform (FFT) is applied along the fast-time dimension to obtain range information. Static-clutter removal is then applied to suppress stationary background components. The virtual array is formed from the transmit–receive antenna channels, after which angle FFT and Doppler FFT operations are used to obtain the radar views required by the downstream perception model. The overall processing flow is summarized in Figure 3.2.

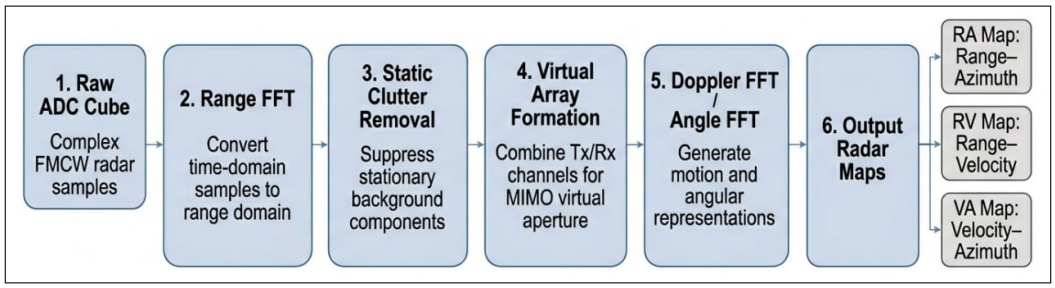


Figure 3.2: Radar signal-processing chain used to generate the range–azimuth, range–velocity, and velocity–azimuth representations from the raw ADC cube.

The RA representation describes spatial target distribution in the range–angle domain. In the proposed two-stage pipeline, this is the only radar view modified by the super-resolution network, because angular resolution is directly limited by the sparse virtual antenna aperture. The RV representation describes range–Doppler characteristics and is generated using standard Doppler processing. The VA representation describes velocity–angle information and is also generated from standard FFT-based processing. Both RV and VA are not super-resolved in this thesis, since velocity resolution is governed by chirp timing and the number of chirps rather than by the number of virtual antenna elements.

Table 3.1: Radar views generated from each ADC cube for the downstream perception pipeline.

View	Physical meaning	Tensor representation
Range–Azimuth (RA)	Spatial distribution in range and angle. Pipeline A uses the raw FFT-based RA representation, while Pipeline B uses the SR-enhanced RA representation.	Complex-valued, real and imaginary channels: $[2, T, 128, 128]$
Range–Velocity (RV)	Doppler signature in range and radial velocity.	Magnitude only: $[1, T, 128, 128]$
Velocity–Azimuth (VA)	Motion and angular distribution in velocity and angle.	Magnitude only: $[1, T, 128, 128]$

After the radar views are generated, each view is normalized to the range $[0, 1]$ and resized to 128×128 for network input. For temporal modelling in the detection stage, consecutive frames are grouped into windows of length $W = 16$ with stride 8. Thus, the radar perception model receives windowed RA, RV, and VA tensors rather than single-frame inputs.

For the SR-enhanced pipeline, the Stage 1 network produces an enhanced RA magnitude map at the native radar resolution. To retain the complex-valued RA representation required by the complex convolutional branch of the detection network, the enhanced magnitude is combined with the phase of the raw FFT-based RA map:

$$X_{\mathrm{RA}}^{\mathrm{SR}} = \left|\hat{X}_{\mathrm{RA}}^{\mathrm{SR}}\right| \exp\left(j \angle X_{\mathrm{RA}}^{\mathrm{raw}}\right), \tag{3.1}$$

where $\left|\hat{X}_{\mathrm{RA}}^{\mathrm{SR}}\right|$ is the magnitude output of the super-resolution network and $\angle X_{\mathrm{RA}}^{\mathrm{raw}}$ is the phase of the raw FFT-based RA representation. This preserves the complex RA input format while injecting the angular enhancement produced by Stage 1. The SR operation is applied before the final 128×128 resizing step so that the learned angular detail is not removed by premature downsampling.

3.3 Stage 1: Radar Super-Resolution

The first stage of the proposed pipeline performs angular super-resolution in the range–azimuth domain. Its purpose is to reconstruct a dense-array-equivalent RA map from a sparse low-cost radar configuration. The network takes the RA representation produced from a sparse $2\mathrm{Tx} \times 4\mathrm{Rx}$ virtual array and learns to approximate the RA representation of a dense $5\mathrm{Tx} \times 10\mathrm{Rx}$ virtual array. This stage therefore addresses the angular-resolution limitation of low-cost automotive radar hardware.

The super-resolution stage is formulated as an image-to-image regression problem in the range–

azimuth domain. The input is a five-channel feature tensor derived from the sparse array, and the target is the dense-array RA magnitude. The output of this stage is later used as the enhanced RA input for the detection and classification pipeline, while the RV and VA branches remain generated through standard FFT-based processing.

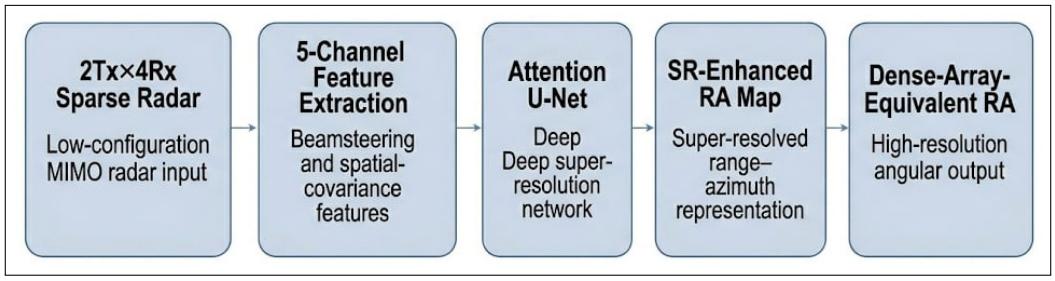


Figure 3.3: Stage 1 radar super-resolution pipeline. The network enhances angular resolution in the range–azimuth domain by mapping sparse-array features to a dense-array-equivalent RA representation.

3.3.1 Sparse-to-Dense Problem Formulation

Let the sparse-array RA feature tensor be denoted by $\mathbf{X}_{\mathrm{LC}}$, where LC refers to the low-configuration radar. The desired target is the dense-array RA representation $\mathbf{Y}_{\mathrm{HC}}$, where HC refers to the high-configuration radar. The super-resolution network learns a nonlinear mapping

$$\hat{\mathbf{Y}}_{\mathrm{HC}} = f_\theta(\mathbf{X}_{\mathrm{LC}}), \tag{3.2}$$

where $f_\theta(\cdot)$ denotes the Attention U-Net with trainable parameters θ, and $\hat{\mathbf{Y}}_{\mathrm{HC}}$ is the reconstructed dense-array-equivalent RA magnitude.

The low-configuration radar uses a $2\mathrm{Tx} \times 4\mathrm{Rx}$ MIMO array, producing 8 virtual antenna elements. The high-configuration target uses a $5\mathrm{Tx} \times 10\mathrm{Rx}$ MIMO array, producing 50 virtual antenna elements. Since angular resolution is determined by the effective virtual aperture, the dense configuration produces sharper angular responses with reduced aliasing and side-lobe effects. The learning task is therefore to infer the angular detail missing from the sparse-array observation.

This formulation is restricted to angular super-resolution. The network does not modify range resolution or Doppler resolution. Range resolution is governed mainly by bandwidth, and velocity resolution is governed by chirp timing and chirp count. Therefore, only the RA branch is enhanced by Stage 1, while RV and VA remain generated by standard radar signal processing.

3.3.2 Five-Channel Input Feature Representation

Instead of using only a single magnitude RA image, the super-resolution network receives a five-channel input tensor. This feature design exposes both beamsteering information and spatial-covariance information to the network. The purpose is to provide richer phase and correlation cues than a magnitude-only representation. The five input channels are listed in Table 3.2.

Table 3.2: Five-channel input representation used by the Stage 1 super-resolution network.

Channel	**Content**
Ch0	Magnitude of the beamsteering response
Ch1	Phase of the beamsteering response
Ch2	Real part of the spatial covariance
Ch3	Imaginary part of the spatial covariance
Ch4	Phase of the spatial covariance

The beamsteering channels describe the directional response of the sparse array, while the covariance channels encode spatial correlation among virtual antenna elements. Each channel is normalized before being passed to the network. This enhanced feature representation follows the radar super-resolution strategy adopted from prior deep-learning-based sparse-array super-resolution work. [7]

3.3.3 Attention U-Net Architecture

The super-resolution network is implemented as an Attention U-Net. The model follows an encoder–decoder structure with skip connections between corresponding resolution levels. The encoder extracts hierarchical spatial features from the sparse-array input, while the decoder reconstructs the enhanced RA map. Attention gates are applied to the skip connections so that the decoder can emphasize spatial regions associated with true targets and suppress clutter, side-lobe responses, and aliasing artifacts inherited from the sparse input.

The network receives a five-channel input tensor and produces a one-channel output, corresponding to the enhanced RA magnitude. The task is treated as continuous regression rather than classification. The final trained model contains $13{,}453{,}837$ trainable parameters.

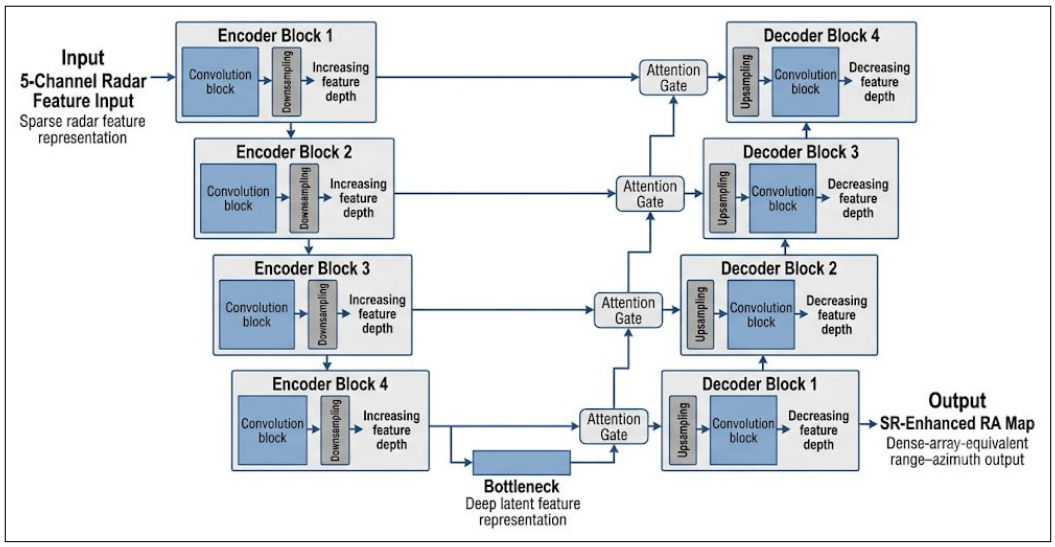


Figure 3.4: Attention U-Net architecture used for Stage 1 radar angular super-resolution.

The regression objective is the mean-squared error between the predicted and target dense-array range–azimuth maps:

$$\mathcal{L}_{\mathrm{SR}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\mathbf{Y}}_{\mathrm{HC}}^{(i)} - \mathbf{Y}_{\mathrm{HC}}^{(i)} \right\|_2^2, \tag{3.3}$$

where $\mathbf{Y}_{\mathrm{HC}}^{(i)}$ is the dense-array target map, $\hat{\mathbf{Y}}_{\mathrm{HC}}^{(i)}$ is the prediction, and N is the batch size.

3.3.4 Dataset3000: Custom Paired Simulated Radar Dataset

A major component of Stage 1 is Dataset3000, a custom paired simulated radar dataset constructed specifically for supervised radar super-resolution. The dataset provides matched low-configuration and high-configuration radar measurements for the same scene. This pairing is essential because the super-resolution network requires an input corresponding to the sparse radar and a target corresponding to the dense radar.

Dataset3000 is generated using a hybrid CARLA–MATLAB procedure. CARLA is used to create the driving scene, including the ego vehicle, road users, their positions, velocities, and the synchronized camera image. However, CARLA's built-in radar sensor is not used to generate the ADC cube. Instead, a custom MATLAB FMCW signal model synthesizes the complex raw radar ADC data analytically from the target list extracted from the CARLA scene. This choice allows the same target configuration to be rendered through both the sparse and dense MIMO radar models, producing perfectly registered low-/high-resolution training pairs.

The main scripts associated with the dataset generation are `collect_full_dataset.m`, which acts as the orchestrator, `collect_radar_classification_dataset.m`, which performs per-frame low- and high-configuration synthesis, and `generateRadarADC`, which implements the FMCW radar signal model. The paired synthesis scripts are used only for

dataset generation and are not included as a separate code appendix in the final document.

For each detected target, the signal model computes its range R, azimuth θ, and radial velocity v. The beat frequency, Doppler frequency, and MIMO spatial phase are modelled as

$$f_b = S\frac{2R}{c}, \tag{3.4}$$

$$f_d = \frac{2vf_c}{c}, \tag{3.5}$$

$$\phi_k = \pi k \sin\theta, \tag{3.6}$$

where S is the FMCW chirp slope, c is the speed of light, f_c is the carrier frequency, and k denotes the virtual antenna element index. The amplitude is determined using a radar-equation-based model that includes transmit power, antenna gains, target radar cross-section, and range-dependent path loss. Additive complex Gaussian noise and ADC quantization are also applied. All targets in a frame are superimposed to form one complex ADC cube.

Both radar configurations are generated from the identical target list for each frame. The low-configuration radar acts as the sparse input, while the high-configuration radar acts as the dense target. Their main parameters are shown in Table 3.3.

Table 3.3: Low- and high-configuration radar models used in Dataset3000.

Parameter	**Low configuration**	**High configuration**
Array	$2\mathrm{Tx} \times 4\mathrm{Rx}$	$5\mathrm{Tx} \times 10\mathrm{Rx}$
Virtual elements	8	50
Carrier frequency	77 GHz	77 GHz
Bandwidth	0.67 GHz	1.50 GHz
Sweep time	60 μs	120 μs
Sample rate	4 MHz	8 MHz
ADC samples	128	256
Chirps	255	512
ADC cube shape	$[128, 255, 4, 2]$	$[256, 512, 10, 5]$
Reference	TI AWR1843	Dense MIMO radar model

The low-configuration radar is chosen to match the TI AWR1843 hardware used for the real radar data in the detection stage. The high-configuration radar is a simulated dense-array model used to generate the dense-array target required for supervised super-resolution training. Figure 3.5 illustrates this low-configuration versus high-configuration setup and highlights why the dense model provides a sharper target representation for the Stage 1 super-resolution network.

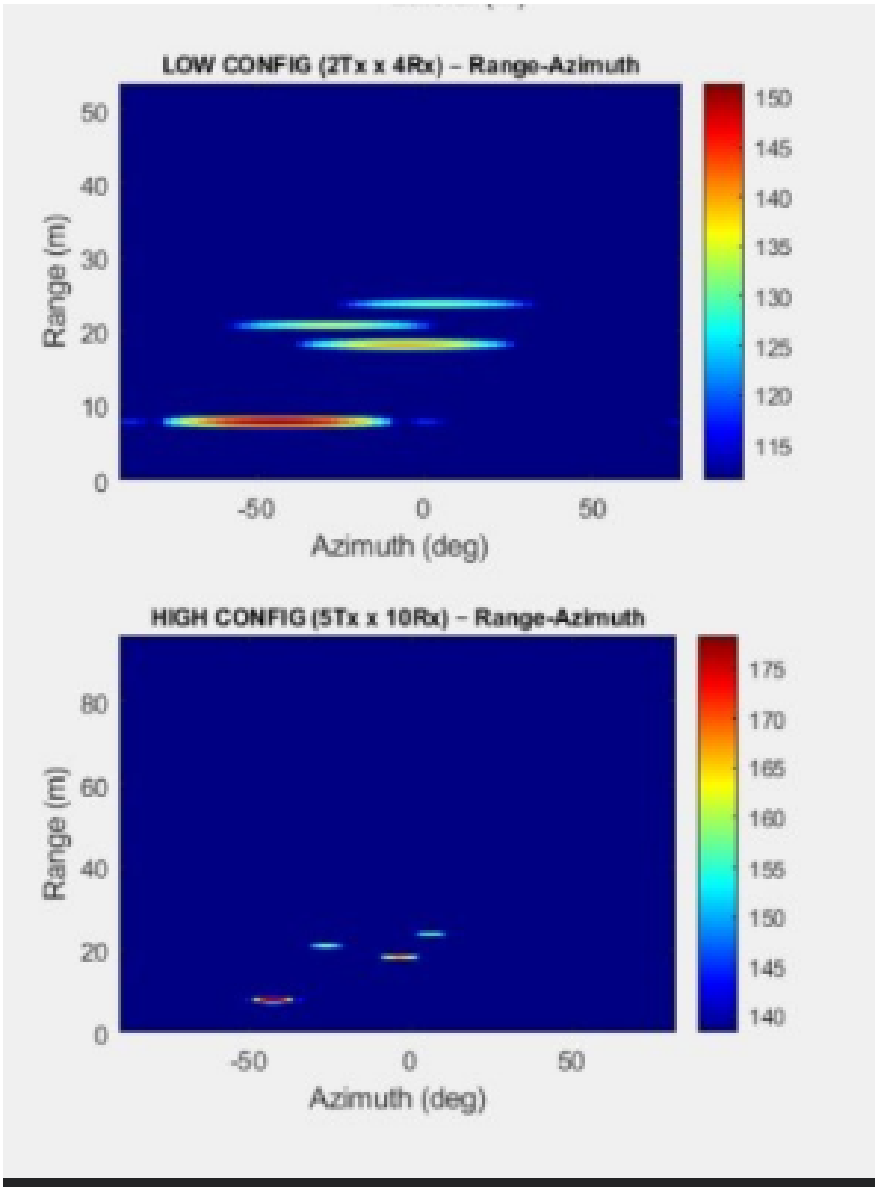


Figure 3.5: Comparison between the low-configuration and high-configuration radar settings used to generate paired sparse and dense samples in Dataset3000.

To improve scene diversity, Dataset3000 is collected across 18 sessions spanning six CARLA maps: Town01, Town02, Town03, Town04, Town05, and Town10HD. These maps include suburban, small-town, urban, motorway/rural, grid-city, and dense photorealistic city environments. The collection also includes three weather conditions: clear, cloudy, and rain/sunset. Multiple ego spawn points are used, and the frame budget is distributed using per-session weights, with 30 frames per scene configuration. A master manifest CSV records the session information.

Each frame contains 1–4 targets placed in front of the ego vehicle, with ranges from 5 m to 25 m and azimuths within $\pm 20^{\circ}$. The currently documented road-user classes are person, cyclist, and car, which correspond to the safety-relevant classes used by the downstream detection and

classification stage. Each class is assigned a representative radar cross-section value in the signal model, with person being the smallest reflector and car being the largest reflector.

The three classes are assigned representative radar cross-section values in the signal model — person 1, cyclist 2, and car 10 (arbitrary linear units) — reflecting their relative reflectivity, with the pedestrian the smallest and the car the largest reflector.

For each frame, the dataset stores the low-configuration ADC cube, the high-configuration ADC cube, the synchronized camera image, and the corresponding label file. The low- and high-configuration radar cubes are stored in `radar_lc/<idx>.mat` and `radar_hc/<idx>.mat`, respectively. The labels are saved as a multi-target CSV containing object identity, class, position, and dimensions. The ADC cubes are complex arrays stored in MATLAB v7.3 HDF5 format. A separate validation script is used to check dataset integrity.

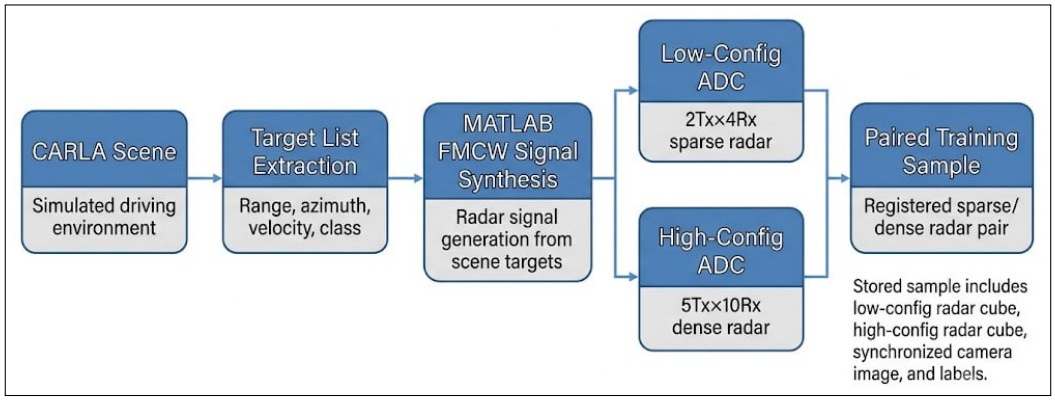


Figure 3.6: Dataset3000 generation workflow. CARLA provides the scene and ground-truth target list, while the custom MATLAB FMCW model synthesizes paired sparse- and dense-array ADC cubes from the same frame.

Dataset3000 is therefore not only a training resource but also a methodological contribution of this thesis. It provides paired sparse/dense radar data that would be difficult to obtain experimentally with perfectly matched target geometry. This enables the supervised learning formulation in Equation (3.2) and provides the foundation for the Stage 1 super-resolution network.

3.4 Stage 2: Detection and Classification

The second stage of the proposed pipeline performs object detection and classification from multi-view radar representations and synchronized camera information. While Stage 1 enhances the range–azimuth representation, Stage 2 uses this representation together with range–velocity, velocity–azimuth, and camera features to detect and classify road users. The retained object classes are person, car, and cyclist.

The Stage 2 model is referred to as RAMP-CNN+. It follows the multi-view radar perception di-

rection of RAMP-CNN/RODNet-style architectures, but extends the pipeline through complex-valued RA processing, camera-guided cross-modal alignment, adaptive multi-view fusion, and a class-balanced asymmetric focal loss. The purpose of this stage is not only to classify the detected road users but also to localize them in the radar grid through heatmap-based object detection.

Two detection pipelines are defined for controlled comparison. Pipeline A uses the raw FFT-based RA map, whereas Pipeline B uses the super-resolution-enhanced RA map generated by Stage 1. In both cases, the RV and VA inputs, ground-truth generation, model architecture, data split, and random seed are kept identical. Therefore, the experimental design is structured to compare raw-RA and SR-enhanced-RA inputs under a controlled setting. However, the detection-stage design and training improvements are attributed to the architectural and training refinements described in this section, not to an unsupported causal claim that super-resolution alone caused the final Stage 2 metric improvement.

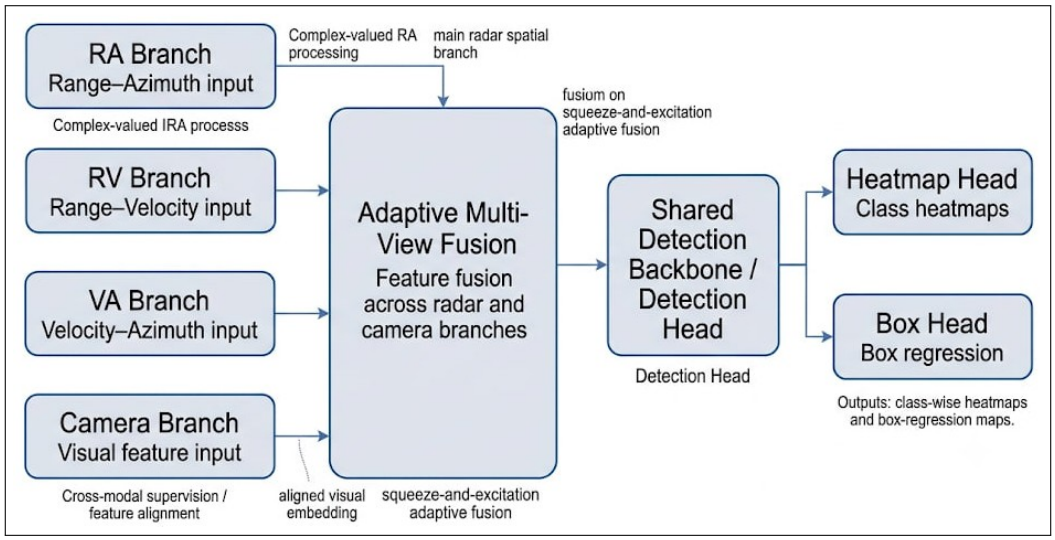


Figure 3.7: Overview of the Stage 2 multi-modal radar object detection and classification pipeline.

3.4.1 Multi-View Inputs

For each radar frame, three radar views are used as input to the detection network: range–azimuth (RA), range–velocity (RV), and velocity–azimuth (VA). These views are generated from the raw ADC cube using the radar signal-processing chain described in Section 3.2. Each radar view is resized to 128×128 before being passed to the network.

The RA view represents the spatial distribution of radar reflections over range and azimuth. In Pipeline B, this view is replaced by the super-resolution-enhanced RA map from Stage 1. Since the complex-valued RA branch requires both magnitude and phase information, the enhanced RA magnitude is combined with the phase of the raw FFT-based RA representation as described

in Equation (3.1). The resulting RA input is represented using real and imaginary channels.

The RV view represents the Doppler response over range and radial velocity. This branch is generated using standard Doppler processing and is not passed through the super-resolution network. The VA view represents velocity and azimuth information and is also generated by standard radar processing. These two views are kept unchanged because the Stage 1 network is designed specifically for angular enhancement in the RA domain.

The camera input is used during training for cross-modal supervision. It provides visual scene information that guides radar feature learning through the cross-modal alignment branch. The input tensor shapes used by the model are summarized in Table 3.4.

Table 3.4: Input and target tensor shapes used in the Stage 2 detection pipeline.

Component	**Shape**	**Description**
RA input	$[2, 16, 128, 128]$	Complex RA representation using real and imaginary channels.
RV input	$[1, 16, 128, 128]$	Magnitude range–velocity representation.
VA input	$[1, 16, 128, 128]$	Magnitude velocity–azimuth representation.
Camera input	$[3, 224, 224]$	Synchronized monocular camera frame used for cross-modal supervision.
Heatmap target	$[3, 16, 128, 128]$	Class-wise ground-truth heatmaps for person, car, and cyclist.
Box target	$[4, 16, 128, 128]$	Box-regression target.
Foreground mask	$[16, 128, 128]$	Mask indicating valid foreground locations.

Temporal information is incorporated by grouping consecutive radar frames into clips of length $W = 16$ with stride 8. This gives the 3-D convolutional model short-term temporal context while preserving the spatial structure of the RA, RV, and VA representations. The resulting numbers of temporal windows for the curated split are shown in Table 3.5.

Table 3.5: Temporal-window counts used for Stage 2 model training and evaluation under the curated $65/20/15$ split.

Split	**Number of windows**
Train	1,353
Validation	467
Test	261

3.4.2 RAMP-CNN+ Architecture

The detection and classification model used in this thesis is RAMP-CNN+. It receives the RA, RV, VA, and camera inputs and produces two main outputs: a class-wise detection heatmap and a box-regression map. The total number of trainable parameters is 36.19 million.

RAMP-CNN+ contains four principal architectural and training contributions. The first is a complex-valued convolutional auto-encoder branch, denoted as CV-CAE, for processing the complex RA input. This branch uses complex 3-D convolution so that phase information in the RA representation is not discarded. The complex RA branch contains approximately 16.5 million parameters.

The second contribution is cross-modal feature alignment (CMFA). A frozen EfficientNet-B0 camera branch produces a 128-dimensional visual embedding, which is aligned with the radar embedding during training. This allows camera supervision to guide radar feature learning.

The third contribution is squeeze-and-excitation adaptive fusion. This module applies class-aware channel attention to fuse the RA, RV, and VA branch features. Since different object classes may be more distinguishable in different radar views, adaptive fusion allows the network to reweight multi-view features according to their relevance.

The fourth contribution is a class-balanced asymmetric focal loss for the heatmap head. This loss is designed for class imbalance and for Gaussian heatmap targets whose peak values approach, but do not exactly equal, one. This is important because the rendered ground-truth peaks reach approximately 0.96, and therefore the positive set is defined using a threshold rather than exact equality to one.

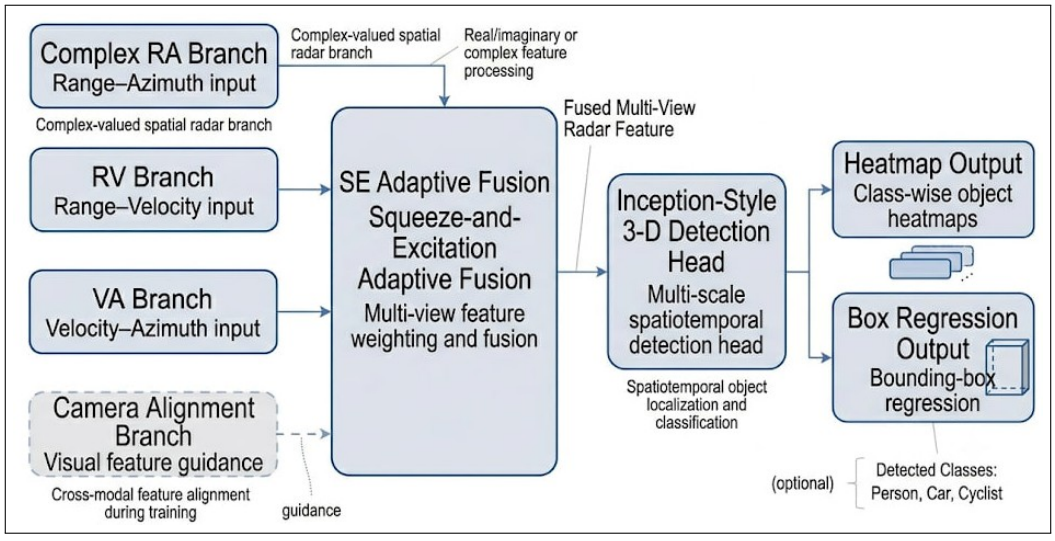


Figure 3.8: RAMP-CNN+ architecture used for multi-view radar object detection and classification.

The final detection head is an Inception-style 3-D head with parallel kernels of sizes $3 \times 3 \times 3$,

$3 \times 5 \times 5$, and a dilated $3 \times 1 \times 21$ angular kernel. This head produces a class-wise heatmap and a box-regression output. The heatmap output has shape $[B, 3, 16, 128, 128]$, where the three channels correspond to person, car, and cyclist. A sigmoid activation is applied to the heatmap output. The box-regression output has shape $[B, 4, 16, 128, 128]$. The heatmap-head bias is initialized to -4.595, giving an initial sigmoid output of approximately 0.01, which acts as a background prior and stabilizes early focal-loss training.

3.4.3 Ground-Truth Heatmap Generation

For each labelled object, the metric centre position is converted into range and azimuth. The dataset label provides the object centre as (p_x, p_y), where p_x denotes lateral position and p_y denotes forward position. The corresponding polar coordinates are computed as

$$r = \sqrt{p_x^2 + p_y^2}, \tag{3.7}$$

and

$$\theta = \tan^{-1}\left(\frac{p_x}{p_y}\right). \tag{3.8}$$

The range and azimuth values are then mapped to the 128×128 radar grid.

The range grid extends from 0.67 m to 27.68 m over 128 bins, corresponding to approximately 0.213 m per bin. The azimuth grid spans from -90° to $+90^\circ$ across 128 angle bins.

A two-dimensional Gaussian peak is rendered at the corresponding range–azimuth location of each object in the heatmap channel of its class. The heatmap therefore has one channel for person, one for car, and one for cyclist. The physical box dimensions from the label file are encoded in the box-regression target, and a foreground mask marks valid object locations.

A critical convention in this thesis is that range is mapped to the heatmap row axis and azimuth is mapped to the heatmap column axis:

$$\text{row} \leftarrow \text{range bin}, \qquad \text{column} \leftarrow \text{angle bin}. \tag{3.9}$$

This convention keeps the ground-truth heatmap spatially consistent with the RA input and with the detection and evaluation logic.

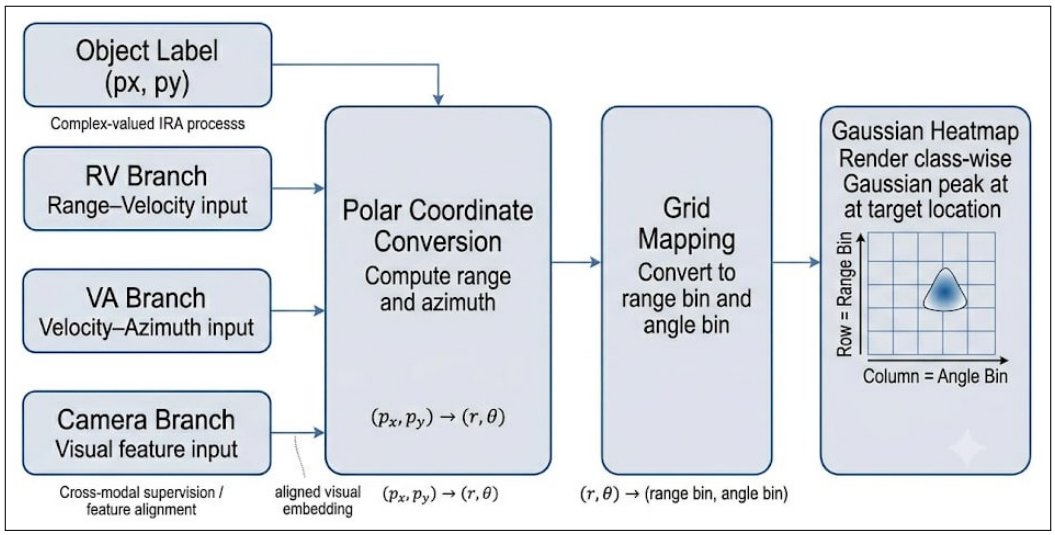


Figure 3.9: Ground-truth heatmap generation for Stage 2 detection. Range is mapped to rows and azimuth is mapped to columns to remain consistent with the RA representation.

The Gaussian peaks reach approximately 0.96 rather than exactly 1.0. Therefore, positive heatmap locations are defined as pixels with ground-truth heatmap value greater than 0.9, while the remaining pixels are treated as negatives for the heatmap loss.

3.4.4 Loss Functions and Training Objective

The Stage 2 training objective combines heatmap detection, box regression, auxiliary radar supervision, and cross-modal alignment. The total loss can be written as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{heatmap}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{aux}} + \mathcal{L}_{\text{align}}, \tag{3.10}$$

where $\mathcal{L}_{\text{heatmap}}$ is the class-balanced asymmetric focal loss, $\mathcal{L}_{\text{box}}$ is the box-regression loss, $\mathcal{L}_{\text{aux}}$ is the auxiliary focal loss applied through the RV/VA-only head, and $\mathcal{L}_{\text{align}}$ is the cross-modal alignment loss between radar and camera embeddings.

The heatmap loss uses class-balanced weighting computed from the retained three-class training set. The weights are 0.88 for person, 1.12 for car, and 1.04 for cyclist. These weights are used to reduce class imbalance while avoiding extreme reweighting. The asymmetric focal formulation emphasizes informative foreground and hard-negative locations while reducing the dominance of background pixels in the heatmap.

The class-balanced asymmetric focal loss is implemented as

$$\mathcal{L}_{\text{heatmap}} = -\frac{1}{N_+} \sum_{b,c,t,i,j} w_c \left[\kappa M^+_{bctij} (1 - \hat{p}_{bctij})^{\alpha_+} \log(\hat{p}_{bctij}) + M^-_{bctij} (1 - y_{bctij})^{\beta} \hat{p}^{\alpha_-}_{bctij} \log(1 - \hat{p}_{bctij}) \right], \tag{3.11}$$

where y_{bctij} is the ground-truth heatmap value, $\hat{p}_{bctij}$ is the predicted heatmap value after clamp-

ing to the range $[10^{-4}, 1-10^{-4}]$, and w_c is the class-balanced weight for class c. The positive and negative masks are defined as

$$M^{+}_{bctij} = \begin{cases} 1, & y_{bctij} > 0.9, \\ 0, & \text{otherwise}, \end{cases} \qquad M^{-}_{bctij} = \begin{cases} 1, & y_{bctij} \leq 0.9, \\ 0, & \text{otherwise}. \end{cases} \tag{3.12}$$

Here, $N_+ = \max(\sum M^{+}_{bctij}, 1)$ is the number of positive heatmap locations used for normalization. The hyperparameters used in the implementation are $\alpha_+ = 2.0$, $\alpha_- = 4.0$, $\beta = 4.0$, and $\kappa = 4.0$.

The full Stage 2 loss combines the main heatmap loss, box-regression loss, auxiliary RV/VA heatmap and box losses, and radar–camera alignment loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{heatmap}} + \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \gamma_{\text{aux}}\left(\mathcal{L}_{\text{aux_heatmap}} + \lambda_{\text{box}}\mathcal{L}_{\text{aux_box}}\right) + \lambda_{\text{align}}\mathcal{L}_{\text{align}}. \tag{3.13}$$

The implementation uses $\lambda_{\text{box}} = 1.0$, $\gamma_{\text{aux}} = 0.5$, and $\lambda_{\text{align}} = 0.1$. The box-regression term is a Smooth-L1 loss evaluated at object-centre mask locations, and the alignment term is based on cosine distance between the radar and camera embeddings.

The optimization configuration used for Stage 2 is summarized in Table 3.6. Mixed precision training is applied using AMP with gradient scaling. Gradient clipping and non-finite-loss guards are included to improve training stability.

Table 3.6: Training configuration for the Stage 2 detection and classification model.

Setting	**Value**
Optimizer	AdamW
Learning rate	1×10^{-4}
Learning-rate schedule	OneCycleLR
Weight decay	1×10^{-4}
Batch size	4
Mixed precision	AMP with GradScaler
Maximum epochs	50
Early stopping	Patience 8, monitoring validation loss
Random seed	42
Gradient clipping	Max-norm 10.0
NaN/inf guard	Skip batch and reset scaler state

Training is performed under Kaggle GPU notebook constraints. To reduce repeated computation, the RA, RV, and VA maps are precomputed once and cached, while the ground-truth heatmaps are generated on the fly. Checkpointing is used so that the trained model is saved

immediately after training and preserved across notebook sessions.

3.5 Stage 2: Prototype-Based Classification Model

In addition to the heatmap-based RAMP-CNN+ detector described in Section 3.4, a second Stage 2 model is developed for classification and object counting. This model is referred to as `FullModel`. Unlike the heatmap-based detector, this model does not produce dense spatial heatmaps or box-regression maps. Instead, it receives the multi-view radar inputs and produces two compact scene-level outputs: class logits and object-count predictions.

The model takes the same radar-view inputs used in the Stage 2 radar pipeline, namely range–azimuth (RA), range–velocity (RV), and velocity–azimuth (VA). The output class logits have shape $[B, 3]$, corresponding to the three retained classes: person, car, and cyclist. The object-count head produces an output of shape $[B, 4]$. The four bins correspond to predicted counts of zero, one, two, and three or more objects in the scene.

The purpose of this model is complementary to the heatmap-based detector. While RAMP-CNN+ focuses on spatial object localization through class-wise heatmaps, the prototype-based model focuses on scene-level classification and object counting. Its design uses physically motivated radar projections and learned class prototypes to obtain a compact representation of the radar scene.

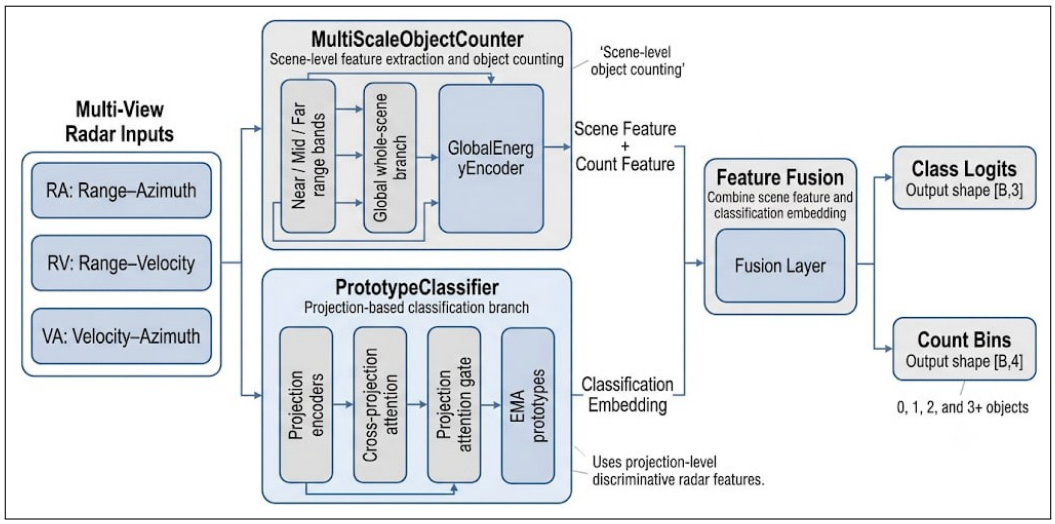


Figure 3.10: Architecture of the prototype-based Stage 2 classification and object-counting model. The model contains a multi-scale object-counting branch and a projection-based prototype-classification branch.

3.5.1 Overall Model Structure

The `FullModel` architecture consists of two parallel branches. The first branch, called `MultiScaleObjec` extracts scene-level information for object counting from the RA and RV middle frames. The second branch, called `PrototypeClassifier`, encodes multiple radar projection views and performs prototype-based class prediction. The scene feature generated by the counting branch is fused with the projection-based embedding before final classification.

Let the radar input views be denoted by $\mathbf{X}_{\mathrm{RA}}$, $\mathbf{X}_{\mathrm{RV}}$, and $\mathbf{X}_{\mathrm{VA}}$. The model produces

$$\mathbf{z}_{\mathrm{cls}} \in \mathbb{R}^{B\times 3}, \qquad \mathbf{z}_{\mathrm{count}} \in \mathbb{R}^{B\times 4}, \tag{3.14}$$

where $\mathbf{z}_{\mathrm{cls}}$ denotes the class-logit output and $\mathbf{z}_{\mathrm{count}}$ denotes the object-count output. Here, B is the batch size.

3.5.2 MultiScaleObjectCounter Branch

The first branch of `FullModel` is the `MultiScaleObjectCounter`. This branch processes the middle frames of the RA and RV inputs. Instead of representing the entire radar scene using only a single global feature, the branch divides the radar maps into three spatial scale bands corresponding to near, mid, and far range thirds. These scale-specific branches allow the model to learn features from different range regions separately.

In addition to the near-, mid-, and far-range branches, the `MultiScaleObjectCounter` includes a global whole-scene branch. The global branch captures information that may not be confined to a single range band. The outputs of the range-band branches and the global branch are fused into a single scene feature vector.

A `GlobalEnergyEncoder` is also used in this branch to extract amplitude-based statistics from the radar maps. These global energy features provide an additional discriminative cue because different object types can produce different radar-return strengths. In particular, the higher radar cross-section of cars can make amplitude statistics useful for distinguishing cars from weaker reflectors. The fused scene feature is then passed to the count head, which predicts the object-count output $\mathbf{z}_{\mathrm{count}}$.

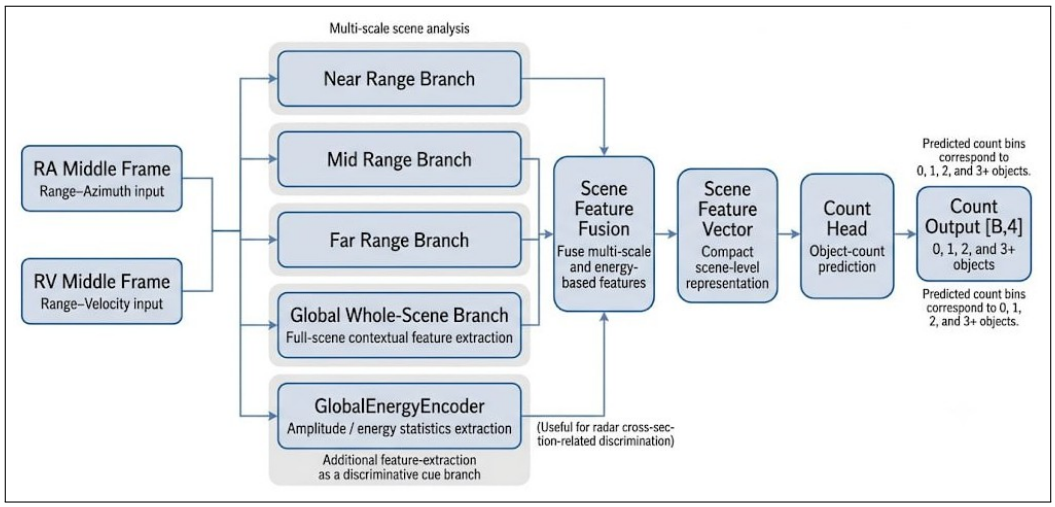


Figure 3.11: MultiScaleObjectCounter branch used for scene-level feature extraction and object-count prediction.

3.5.3 PrototypeClassifier Branch

The second branch of `FullModel` is the `PrototypeClassifier`. This branch is designed around the physical intuition that different radar projections contain different discriminative information. Rather than using only the original two-dimensional radar maps, the model encodes six projection views:

$$\{\text{ra_range},\ \text{ra_azimuth},\ \text{rv_range},\ \text{rv_velocity},\ \text{va_azimuth},\ \text{va_velocity}\}. \tag{3.15}$$

Each projection view is encoded using a separate `ProjEncoder2D` module. These encoders extract temporal mean and temporal variance features, allowing the model to represent both the average projection structure and the temporal variation of the radar response. The six encoded projection features are then processed using a `CrossProjAttn` module, which applies cross-projection attention between paired views. This allows the model to learn relationships between complementary radar projections instead of treating each projection independently.

A `ProjectionAttentionGate` applies class-specific weighting to the projection features. This is useful because the most informative radar projection may differ across object classes. For example, velocity-related projections can capture micro-Doppler spread, which is useful for distinguishing cyclists from cars. Azimuth-related projections can capture angular extent, which is useful for representing spatial spread and localization cues. Cross-projection coupling allows the model to represent joint spatial–velocity signatures that may not be visible from a single projection alone.

The resulting projection embedding is fused with the scene feature generated by the `MultiScaleObjectCo` branch. Classification is then performed using both learned exponential-moving-average (EMA)

prototype vectors and a direct classification head. The EMA prototypes provide class-level reference embeddings that are updated smoothly during training. This prototype mechanism is intended to stabilize the classifier under class imbalance by maintaining more consistent class representations.

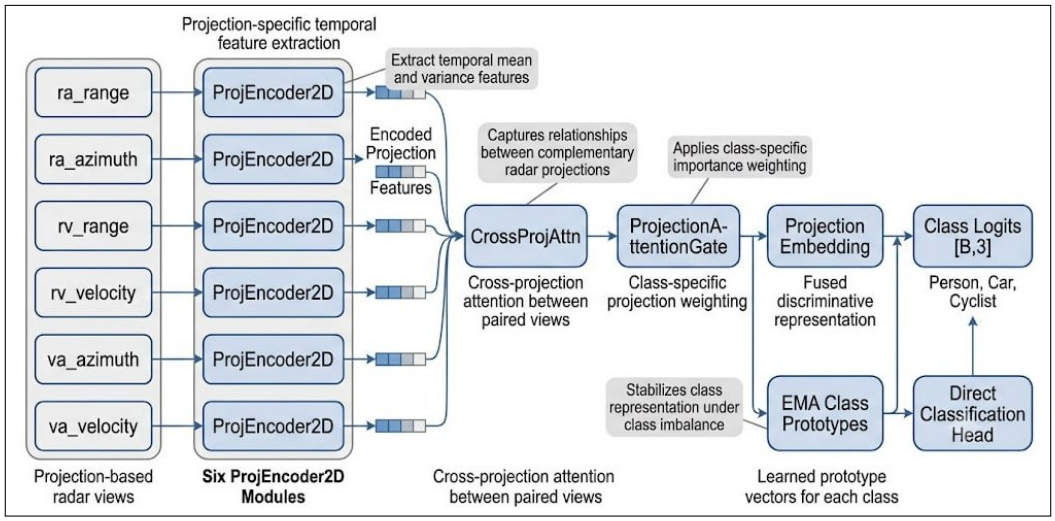


Figure 3.12: PrototypeClassifier branch using projection-based radar encodings, cross-projection attention, class-specific projection gating, and EMA class prototypes.

3.5.4 Training Configuration

The prototype-based classification model is trained using the AdamW optimizer with a learning rate of 5×10^{-5}. The batch size is 8, the dropout probability is 0.30, and the maximum number of training epochs is 60. The training configuration is summarized in Table 3.7.

Table 3.7: Training configuration for the prototype-based Stage 2 classification model.

Setting	**Value**
Model name	`FullModel`
Optimizer	AdamW
Learning rate	5×10^{-5}
Batch size	8
Dropout	0.30
Maximum epochs	60
Classification output	$[B, 3]$ class logits
Count output	$[B, 4]$ object-count prediction

The prototype-based model is optimized using a combined classification, counting, and supervised-contrastive objective. The class-logit output is supervised using a class-weighted focal cross-entropy loss with label smoothing. For a sample with class label y, the smoothed target distri-

bution is defined as

$$q_c = \begin{cases} 1-\epsilon, & c = y, \\ \dfrac{\epsilon}{C-1}, & c \neq y, \end{cases} \tag{3.16}$$

where $C = 3$ is the number of retained classes and $\epsilon = 0.15$ is the label-smoothing factor. The focal classification loss is then written as

$$\mathcal{L}_{\text{cls}} = \frac{1}{B} \sum_{n=1}^{B} w_{y_n} \left(1 - p_{t,n}\right)^{\gamma} \left[- \sum_{c=1}^{C} q_{n,c} \log p_{n,c} \right], \tag{3.17}$$

where $p_{n,c}$ is the softmax probability for class c, $p_{t,n}$ is the probability term derived from the cross-entropy value, w_{y_n} is the class weight for the ground truth class, and $\gamma = 1.5$.

The object-count output is supervised using a standard cross-entropy loss over four count bins corresponding to zero, one, two, and three or more objects:

$$\mathcal{L}_{\text{count}} = \text{CE}\left(\mathbf{z}_{\text{count}}, y_{\text{count}}\right). \tag{3.18}$$

A supervised contrastive loss is also applied to the normalized embedding features to encourage samples from the same class to remain close in the embedding space. The temperature used for this loss is $\tau = 0.07$. The overall training objective is

$$\mathcal{L}_{\text{FullModel}} = \mathcal{L}_{\text{cls}} + 0.20\, \mathcal{L}_{\text{count}} + 0.25\, \mathcal{L}_{\text{supcon}}. \tag{3.19}$$

The EMA class prototypes are updated during training with smoothing factor $\alpha = 0.998$. These prototypes stabilize the class-level embedding representation, but they are updated through the EMA mechanism rather than through a separate explicit prototype-loss term.

3.5.5 Design Rationale

The main design motivation behind the prototype-based model is to exploit the physical meaning of different radar projections. Range, azimuth, and velocity projections do not carry identical information. Velocity projections can describe motion spread and micro-Doppler-like behaviour, which is useful for separating moving road users such as cyclists from larger vehicles. Azimuth projections describe angular spread and spatial extent, which are useful for understanding how targets occupy the radar field of view. Range projections provide complementary distance-dependent structure. By encoding these views separately and then coupling them through cross-projection attention, the model can learn joint radar signatures that are more informative than any single projection alone.

The EMA prototype mechanism provides an additional stabilizing structure for classification.

Instead of relying only on a direct classification head, the model also compares the fused embedding with learned class prototypes. These prototypes act as class-level reference vectors and are updated smoothly during training. This design is particularly useful when the class distribution is not perfectly balanced, because it encourages more stable class representations across training iterations.

Overall, the prototype-based Stage 2 model provides a complementary classification and counting pathway beside the heatmap-based detector. It does not replace the spatial detection model described in Section 3.4; rather, it serves as a second radar-based model that emphasizes projection-level discriminative features, prototype-stabilized classification, and scene-level object counting.

3.6 Dataset and Splits

The Stage 2 detection and classification experiment uses an AWR1843 mmWave FMCW radar dataset with synchronized monocular camera images. The dataset contains 22 driving sequences captured in open parking-lot and road-scene environments. Each radar frame is stored as a complex ADC cube with shape $128 \times 255 \times 4 \times 2$, corresponding to ADC samples, chirps, receive antennas, and transmit antennas. The 4 receive antennas and 2 transmit antennas form an 8-element virtual array under TDM-MIMO operation.

Each frame has a corresponding CSV label file without a header. The six label columns are `uid`, `cls`, `px`, `py`, `wid`, and `len`. Here, `px` and `py` denote the lateral and forward centre position in metres, while `wid` and `len` denote the physical width and length of the object in metres.

3.6.1 Three-Class Setup

Although the original dataset contains more than the three classes retained in this thesis, the final detection and classification task is restricted to person, car, and cyclist. The remaining low-frequency or unused categories are excluded because they contain too few instances for stable three-class training and evaluation. The retained classes and their label mappings are shown in Table 3.8.

Table 3.8: Retained classes for Stage 2 detection and classification.

Class	Original label ID	Model index
Person	0	0
Car	2	1
Cyclist	80	2

The training-set class distribution after this reduction is determined from the manual curated $65/20/15$ sequence-level split. In the training set, person accounts for 44.5% of the retained instances, car accounts for 26.0%, and cyclist accounts for 29.5%. Thus, the three-class setup is moderately imbalanced, with car having the lowest number of training instances among the retained classes.

3.6.2 Sequence-Level Split

The dataset is split at the sequence level rather than at the frame level. This prevents temporal leakage, because frames from the same driving sequence do not appear in more than one split. A manual curated $65/20/15$ train/validation/test split is used, with random seed 42 for reproducibility.

The split is designed so that the training set provides the model with representative examples of the retained classes, while the validation and test sets remain sequence-wise independent. This is important for radar perception because adjacent frames within the same driving sequence are highly correlated. A frame-level random split could therefore produce overly optimistic validation or test performance by allowing temporally similar frames from the same scene to appear in both training and evaluation sets.

Table 3.9: Labelled-frame counts for the manual curated $65/20/15$ sequence-level split.

Split	**Sequences**	**Labelled frames**
Train	13	10,832
Validation	5	3,744
Test	4	2,100
Total	22	16,676

The per-class instance counts for the curated split are given in Table 3.10. These counts are used to examine the class balance of the detection problem and to justify the class-balanced heatmap loss described in Section 3.4.4.

Table 3.10: Per-class instance counts in the manual curated $65/20/15$ Stage 2 split.

Class	**Train**	**Validation**	**Test**
Person	6,819	2,460	1,376
Car	3,978	1,665	927
Cyclist	4,526	2,847	867

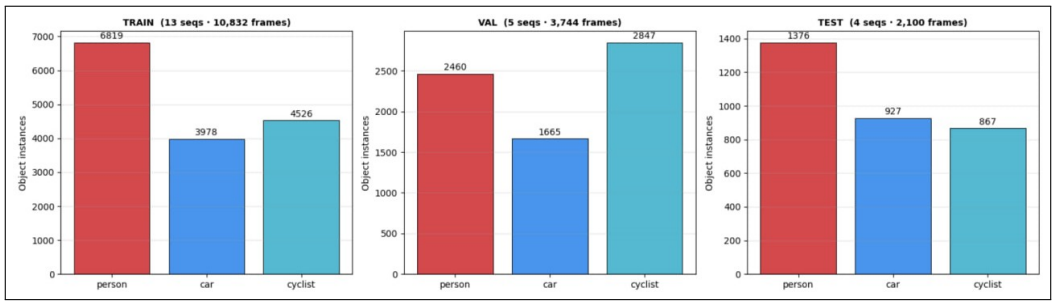


Figure 3.13: Dataset statistics for the retained three-class Stage 2 detection problem.

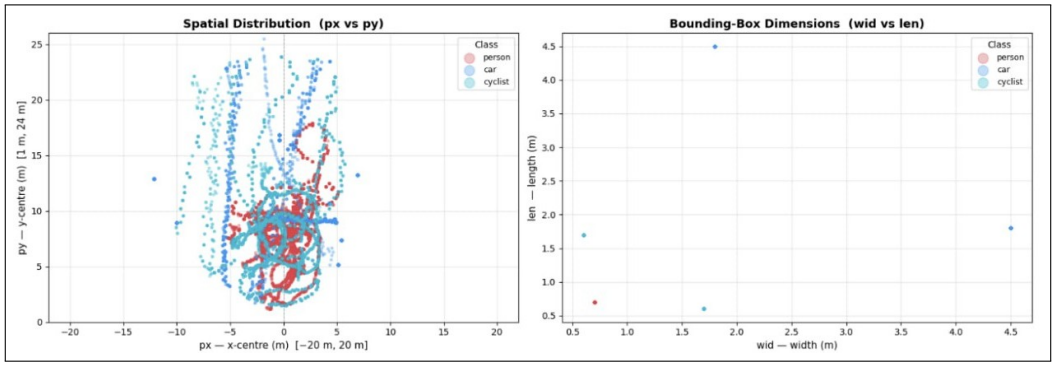


Figure 3.14: Bounding-box and spatial statistics for the retained Stage 2 classes.

3.7 Design Considerations

The proposed two-stage radar perception pipeline involves several engineering design tradeoffs. These tradeoffs are important because the system is intended for an autonomous-driving perception setting, where hardware cost, sensing resolution, computational feasibility, and reliability must be considered together. This section summarizes the major design considerations addressed in the methodology: cost versus angular resolution, compute and memory constraints, real-time feasibility, and dataset and simulation limitations.

3.7.1 Cost versus Angular Resolution

A central motivation of this thesis is the tradeoff between radar hardware cost and angular resolution. In a MIMO radar system, angular resolution is strongly related to the effective virtual aperture, which depends on the number and placement of virtual antenna elements. A dense antenna configuration can produce sharper angular responses, but it requires more transmit and receive channels, increasing hardware complexity and cost. This is a significant constraint for low-cost automotive radar deployment.

The Stage 1 super-resolution module is designed to address this tradeoff. Instead of directly relying on a dense radar array at deployment, the network learns a mapping from a sparse

$2\mathrm{Tx} \times 4\mathrm{Rx}$ configuration to a dense-array-equivalent $5\mathrm{Tx} \times 10\mathrm{Rx}$ range–azimuth representation. The sparse configuration contains 8 virtual antenna elements, while the dense target configuration contains 50 virtual antenna elements. This design choice reflects the practical engineering objective of improving effective angular detail while keeping the physical radar hardware close to a low-cost configuration.

However, this approach does not remove the physical limits of radar sensing. The super-resolution network enhances only the angular structure of the range–azimuth representation. It does not improve range resolution or velocity resolution, since those are governed by bandwidth, chirp timing, and chirp count rather than by the number of antenna elements. Therefore, the proposed design applies super-resolution only to the RA branch, while the RV and VA representations are obtained through standard radar signal processing as described in Section 3.2. This separation keeps the design physically interpretable and prevents overclaiming the scope of the super-resolution stage.

3.7.2 Compute and Memory Constraints

The methodology is also constrained by available computational resources. Both stages of the pipeline involve deep neural networks and high-dimensional radar tensors. Stage 1 uses an Attention U-Net with $13{,}453{,}837$ trainable parameters, while Stage 2 uses RAMP-CNN+ with approximately 36.19 million trainable parameters. In addition, Stage 2 uses temporal radar clips of length $W = 16$, with RA, RV, and VA maps resized to 128×128. These choices improve modelling capacity but increase memory demand.

The experiments were conducted under Kaggle GPU notebook constraints, including limited session duration and weekly GPU quota. This influenced several implementation decisions. For Stage 2, the RA, RV, and VA maps are precomputed and cached instead of being recomputed during every training epoch. Ground-truth heatmaps are generated during training, while the radar views are loaded from cached files. This reduces repeated signal-processing overhead and makes training feasible within the available GPU time.

Memory constraints also influenced the training configuration. A batch size of 4 is used for Stage 2 to fit the multi-view temporal tensors and the RAMP-CNN+ model within GPU memory. Mixed precision training with AMP and gradient scaling is used to reduce memory usage and improve computational efficiency. Gradient clipping and non-finite-loss guards are included to improve training stability. These engineering choices do not change the scientific objective of the model, but they are necessary for obtaining a stable and reproducible implementation under limited compute resources.

Table 3.11: Major engineering constraints and design responses in the proposed methodology.

Design issue	**Constraint**	**Methodological response**
Angular resolution	Sparse low-cost radar arrays have limited angular aperture.	Use Stage 1 super-resolution to enhance the RA representation from a $2\mathrm{Tx} \times 4\mathrm{Rx}$ input toward a dense-array-equivalent target.
Hardware cost	Dense MIMO arrays require more antenna channels and higher system complexity.	Use a low-configuration radar as the practical input and learn angular enhancement in software.
GPU memory	Multi-view temporal tensors and a 36.19 million-parameter detector are memory-intensive.	Use batch size 4, mixed precision training, and cached radar views.
Training time	Kaggle GPU sessions impose limited runtime and quota.	Precompute RA/RV/VA maps, use checkpointing, and apply early stopping.
Physical scope	SR is trained for angular enhancement, not range or Doppler enhancement.	Apply SR only to the RA branch and keep RV/VA generated by standard FFT-based processing.

3.7.3 Real-Time Feasibility

Although this thesis focuses on methodology and experimental validation, real-time feasibility is an important design consideration for autonomous-driving applications. A practical perception pipeline must process radar frames within a limited latency budget. The proposed pipeline contains both conventional radar signal processing and neural network inference. The FFT-based range, Doppler, and angle processing steps are standard operations that are suitable for efficient implementation. The more demanding components are the Stage 1 Attention U-Net and the Stage 2 RAMP-CNN+ detector.

In the present implementation, the radar views are precomputed for training efficiency. This is appropriate for offline experimentation, but a deployed system would need to compute the RA, RV, and VA representations online from incoming ADC frames. Similarly, the Stage 1 super-resolution network and Stage 2 detector would need to run sequentially or in an optimized pipeline. Therefore, the current work should be interpreted as a functional research prototype rather than a fully optimized real-time embedded system.

Several steps would be required before deployment on an automotive platform. First, the end-

to-end inference time should be measured on the target hardware. Second, the Stage 1 and Stage 2 networks may need to be optimized through model compression, pruning, quantization, or lighter backbone designs. Third, the system should be tested under continuous streaming conditions rather than only on cached frames. These steps are left as future engineering work.

3.7.4 Dataset and Simulation Limitations

The dataset design also involves important limitations. Stage 1 is trained using paired simulated data generated from CARLA scenes and a custom MATLAB FMCW signal model. This approach is necessary because perfectly registered sparse- and dense-array measured radar pairs are difficult to obtain experimentally. By synthesizing both array configurations from the same target list, Dataset3000 provides exact low-/high-resolution supervision for the super-resolution task.

However, simulation cannot fully reproduce all real-world radar effects. Material properties, multipath, near-field scattering, sensor imperfections, and environmental clutter may differ between the simulated signal model and measured radar data. The qualitative real-world validation of Stage 1 demonstrates that the trained network can produce cleaner angular peaks on measured radar frames, but quantitative PSNR/SSIM evaluation on real measured dense-array ground truth is limited by the absence of such ground truth.

The Stage 2 detector is evaluated on an AWR1843 mmWave radar dataset with synchronized camera data. The final task is restricted to three retained classes: person, car, and cyclist. Other low-frequency or unused categories are excluded to obtain a stable three-class detection problem. This improves training consistency, but it also limits the scope of the detector. The resulting model should therefore be interpreted as a three-class road-user detector rather than a complete all-object automotive perception system.

The dataset scenes are also largely open parking-lot and road-scene environments. As a result, the trained detector may not fully represent dense urban traffic, complex occlusion, or highly cluttered roadside conditions. Broader evaluation across additional locations, weather conditions, traffic densities, and sensor setups would be required to make stronger claims about general deployment.

Finally, the relationship between Stage 1 super-resolution and Stage 2 detection must be framed carefully. Stage 1 provides evidence of angular enhancement through its own image-quality metrics and qualitative real-world examples. Stage 2 provides evidence of multi-view detection and classification using the architecture and training design described in Section 3.4. In this thesis, the RAMP-CNN+ detector is evaluated using both raw RA input and SR-enhanced RA input under the same evaluation protocol. However, this comparison is still limited to the implemented dataset, model, and array-geometry setting. Broader raw-versus-SR evalu-

ation across additional scenes, radar geometries, and repeated training runs remains a future extension. Therefore, this thesis presents the system as a proposed two-stage radar perception pipeline while interpreting the documented detection-stage behaviour within the implemented methodology and evaluation setting.

3.7.5 Summary of Engineering Design Considerations

Overall, the proposed methodology balances sensing performance, hardware cost, and computational feasibility. The SR stage addresses the cost–resolution tradeoff by learning angular enhancement from sparse-array radar data. The detection stage uses multi-view radar representations and camera-guided supervision to improve radar feature learning for three-class object detection. Compute limitations are handled through precomputation, caching, mixed precision training, small-batch optimization, and checkpointing. At the same time, the methodology recognizes the limitations of simulation-based supervision, restricted class coverage, and offline inference. These considerations define both the engineering value of the proposed pipeline and the boundaries within which its results should be interpreted.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the experimental results of the proposed two-stage radar perception pipeline. The first part evaluates the Stage 1 radar super-resolution model using image-quality metrics and qualitative real-world examples. The second part evaluates the Stage 2 models using radar-based classification and detection metrics on the AWR1843 radar dataset. The discussion then interprets the observed recall–precision behaviour, class-wise performance differences, and dataset-dependent limitations.

4.1 Experimental Setup

The experiments were conducted under Kaggle GPU notebook constraints. These constraints included limited session duration and weekly GPU quota, which influenced the training and evaluation workflow described in Section 3.7. For Stage 1, paired sparse- and dense-array radar representations were used to train and evaluate the super-resolution network. For Stage 2, pre-computed RA, RV, and VA radar views were used with the detection and classification models to reduce repeated signal-processing overhead.

Two groups of evaluation metrics were used. Stage 1 was evaluated using image-quality metrics, since the task is formulated as range–azimuth image reconstruction. Stage 2 was evaluated using object-detection and classification metrics, since the task is to detect, classify, and count road users from radar representations.

4.1.1 Stage 1 Evaluation Metrics

The Stage 1 super-resolution network reconstructs a dense-array-equivalent range–azimuth map from a sparse-array input. Therefore, the predicted RA image is compared with the dense-array target using peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM).

The mean-squared error (MSE) between the predicted image $\hat{I}$ and the reference image I is given by

$$\mathrm{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(I_i - \hat{I}_i \right)^2 , \tag{4.1}$$

where N is the number of pixels. PSNR is then computed as

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{I_{\max}^2}{\mathrm{MSE}} \right) , \tag{4.2}$$

where $I_{\max}$ is the maximum possible image intensity. A higher PSNR indicates lower pixel-level reconstruction error. SSIM measures structural similarity between two images, considering luminance, contrast, and structure. In this thesis, PSNR and SSIM are used together because PSNR captures pixel-level fidelity, while SSIM captures structural consistency in the reconstructed RA map.

4.1.2 Stage 2 Evaluation Metrics

The heatmap-based Stage 2 detector is evaluated using precision, recall, F1 score, mean average precision (mAP), and mean average recall (mAR). For a given class, precision is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{4.3}$$

where TP denotes true positives and FP denotes false positives. Recall is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{4.4}$$

where FN denotes false negatives. The F1 score is the harmonic mean of precision and recall:

$$\text{F1} = \frac{2\,\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4.5}$$

For the centre-frame detection evaluation, predicted heatmap peaks are matched against ground-truth peak centres. A predicted detection is counted as a true positive if it has the same class as the ground-truth object, belongs to the same temporal frame, and its predicted peak location is within the matching tolerance of the ground-truth peak centre. The implementation uses separate range-bin and angle-bin tolerances of $r_{\text{tol}} = 4$ and $a_{\text{tol}} = 4$, giving an effective Euclidean pixel-distance threshold of

$$d_{\text{tol}} = \sqrt{r_{\text{tol}}^2 + a_{\text{tol}}^2} = \sqrt{4^2 + 4^2} \approx 5.66 \text{ bins}. \tag{4.6}$$

Each ground-truth object is matched at most once, using predictions sorted by confidence. Predictions that do not match any ground-truth object are counted as false positives, while ground-truth objects that are not matched by any prediction are counted as false negatives.

The mAP value summarizes average precision over the evaluated classes, while mAR summarizes average recall over the evaluated classes. These aggregate metrics provide a single summary of detection performance across the retained classes: person, car, and cyclist.

The prototype-based Stage 2 classification model is evaluated separately because it produces scene-level class logits and object-count predictions rather than dense detection heatmaps. Its results are therefore reported using classification and counting metrics rather than heatmap-

detection metrics.

The prototype-based Stage 2 model is evaluated using training accuracy, validation accuracy, training loss, validation loss, and row-normalized confusion matrices. Since `FullModel` produces scene-level class logits rather than dense heatmap detections, its results are reported separately from the RAMP-CNN+ detection metrics. The object-counting branch is included in the model formulation, but separate count accuracy or count-error metrics are not reported in this thesis because the final evaluation figure focuses on classification behaviour.

4.2 Stage 1 Results

The Stage 1 super-resolution model was trained to reconstruct the dense-array range–azimuth representation from the sparse-array input. The model uses an Attention U-Net with a five-channel input representation and a one-channel enhanced RA magnitude output, as described in Section 3.3.

4.2.1 Training Behaviour

The Stage 1 network was trained for 40 epochs. The best validation performance was obtained at epoch 39. The recorded best-epoch log is shown below:

```
Epoch 39/40 (645.0s) | LR=1.15e-06
  Train loss : 0.000051
  Val   loss : 0.000081   PSNR=42.27 dB   SSIM=0.9463   BEST
  New best model -> /kaggle/working/checkpoints/best_model.pt
```

The training and validation losses were both low, and the small gap between them indicates that the model converged without severe overfitting. The late best epoch also suggests that the model continued improving through most of the training process.

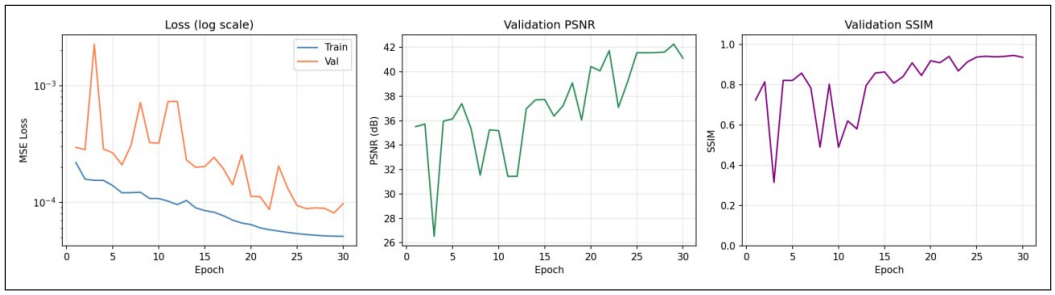


Figure 4.1: Training behaviour of the Stage 1 Attention U-Net super-resolution model.

4.2.2 Quantitative Super-Resolution Results

The quantitative Stage 1 results are summarized in Table 4.1. The validation set achieved a PSNR of 42.27 dB and an SSIM of 0.9463 at the best epoch. On the held-out test set, the model achieved a PSNR of 40.10 dB and an SSIM of 0.8716, with a test loss of 0.006973.

Table 4.1: Quantitative results of the Stage 1 radar super-resolution model.

Split	**PSNR (dB)**	**SSIM**	**Loss**
Validation	42.27	0.9463	0.000081
Test	40.10	0.8716	0.006973

The test PSNR above 40 dB and the SSIM above 0.87 indicate that the network reconstructs the dense-array RA representation with high fidelity. PSNR reflects the low pixel-level reconstruction error, while SSIM confirms that the reconstructed output preserves the structural pattern of the dense-array target.

4.2.3 Qualitative Sim-to-Real Validation

In addition to quantitative evaluation on data with dense-array targets, the super-resolution model was qualitatively tested on independently measured real-world radar data. This test is important because the Stage 1 model is trained using simulated and synthetic paired data, whereas real measured dense-array ground truth is not available.

The qualitative comparison shows that the raw sparse-array delay-and-sum RA output is strongly affected by angular aliasing and side-lobe artifacts. In contrast, the super-resolution output produces cleaner and more focused angular peaks. These peaks align with objects visible in the synchronized camera frame. This qualitative result supports the sim-to-real transfer of the Stage 1 model as an angular enhancement module, while the quantitative PSNR/SSIM values remain limited to settings where dense-array targets are available.

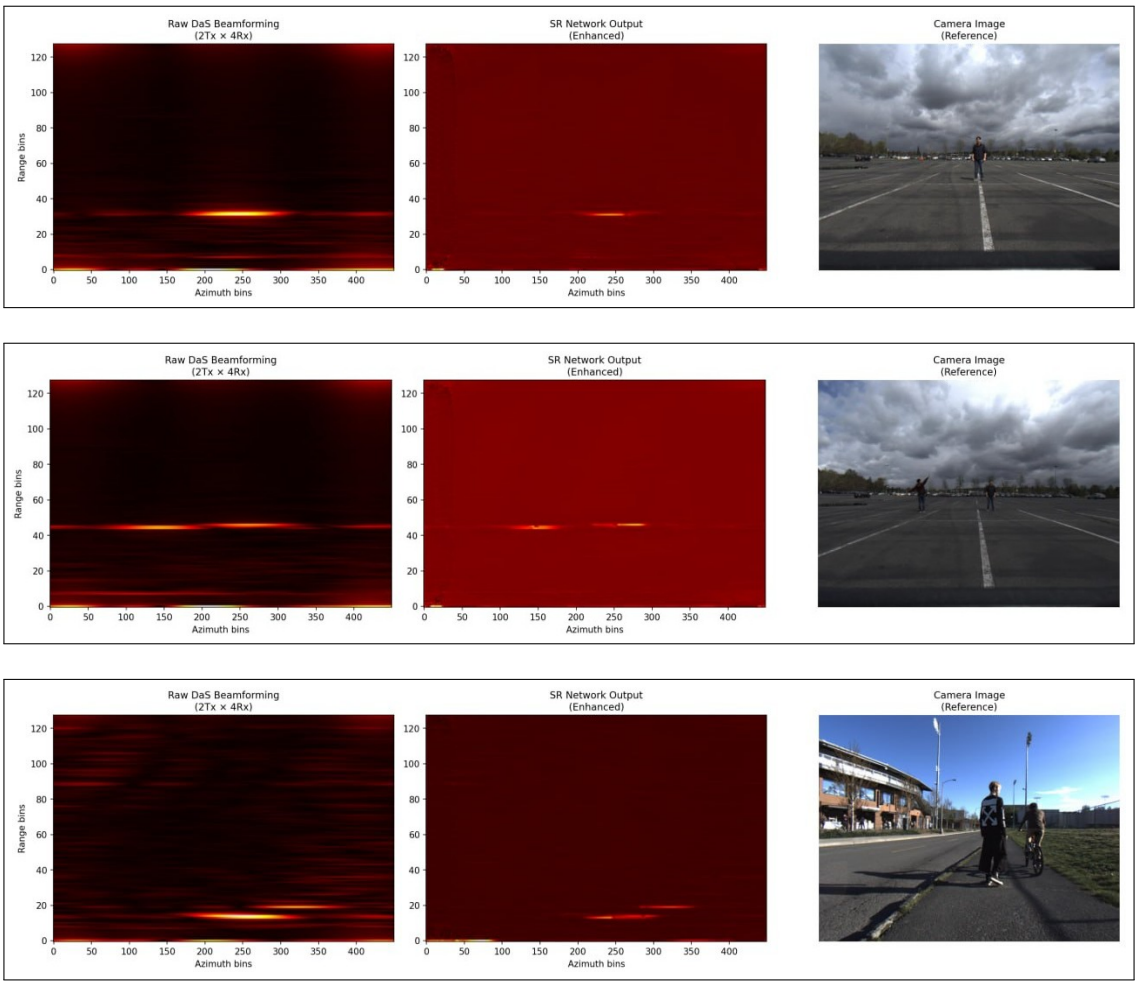


Figure 4.2: Qualitative sim-to-real validation of the Stage 1 super-resolution model. The SR output should be compared with the raw sparse-array RA map and the synchronized camera frame.

4.3 Stage 2 Results

The Stage 2 experiments evaluate radar-based perception on the retained three-class road-user problem: person, car, and cyclist. Two Stage 2 models are considered. The first is RAMP-CNN+, a heatmap-based detection and classification model. The second is `FullModel`, a prototype-based classification and object-counting model. The RAMP-CNN+ results are reported using detection metrics, while the prototype-based model is reported separately because its outputs are class logits and count bins rather than dense detection heatmaps.

4.3.1 Dataset Statistics

The Stage 2 dataset consists of AWR1843 radar sequences with synchronized camera images and frame-level object labels. The retained three-class setup and sequence-level split are described in Section 3.6. Dataset statistics are important for interpreting the class-wise results

because class frequency and scene composition affect both precision and recall.

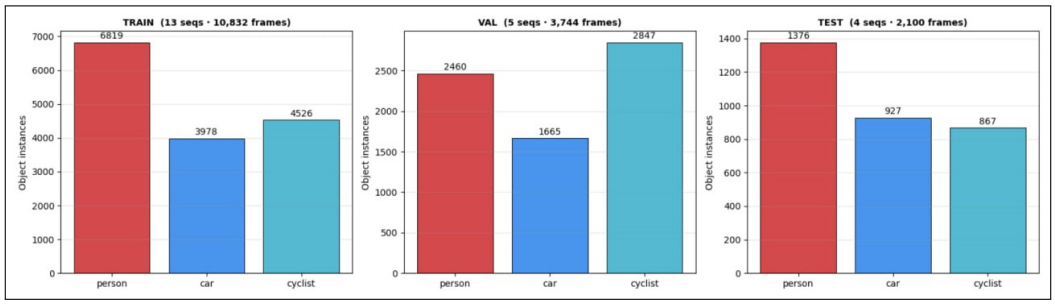


Figure 4.3: Dataset statistics for the manual curated $65/20/15$ Stage 2 three-class detection and classification split.

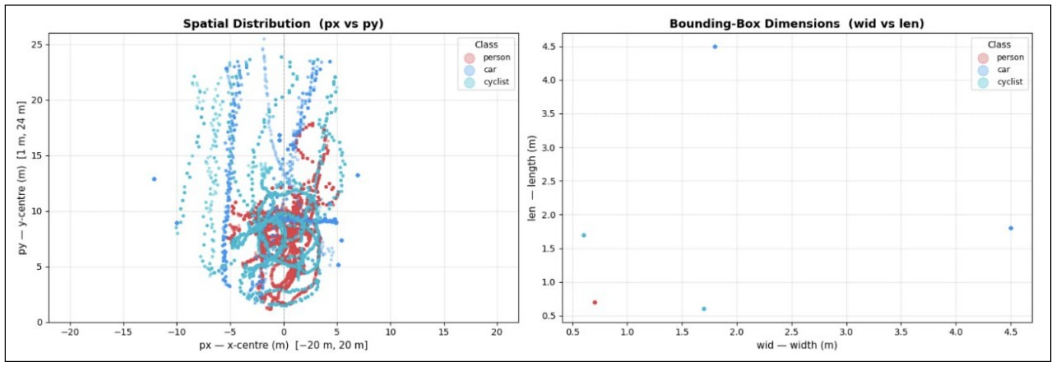


Figure 4.4: Bounding-box and spatial statistics for the Stage 2 dataset.

4.3.2 RAMP-CNN+ Training Behaviour

The RAMP-CNN+ training process used the training configuration described in Section 3.4.4. The final run stopped by early stopping after 21 epochs, with the best validation loss of 5.1517 obtained at epoch 13. The per-epoch training time was approximately 900–1040 s on a single GPU.

During training, the focal-loss component decreased from approximately 16.5 to approximately 0.3. The gradient norms showed large early spikes, around 100, and then settled to single or low-double digit values. Skipped batches became near-zero after the first epoch. This behaviour indicates that the stability mechanisms in the training loop, including mixed precision handling, non-finite-loss checks, and gradient clipping, contributed to a stable final training run.

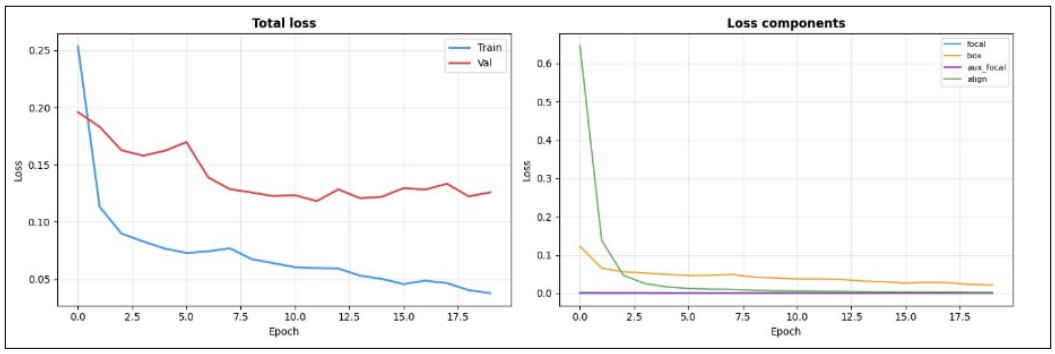


Figure 4.5: Training curves of the RAMP-CNN+ detection and classification model.

4.3.3 RAMP-CNN+ Detection and Classification Performance

The RAMP-CNN+ detector was evaluated in two configurations: without Stage 1 super-resolution and with the SR-enhanced RA representation. The same evaluation criterion was used in both cases so that the raw-RA and SR-enhanced configurations could be compared under the same detection protocol.

Before evaluating the SR-enhanced configuration, the RAMP-CNN+ detector was first evaluated using the raw FFT-based RA representation without the Stage 1 super-resolution input. This configuration provides the baseline detection performance of the heatmap-based detector when only the standard radar signal-processing output is used. The corresponding training curves and normalized confusion matrix are shown in Figure 4.6.

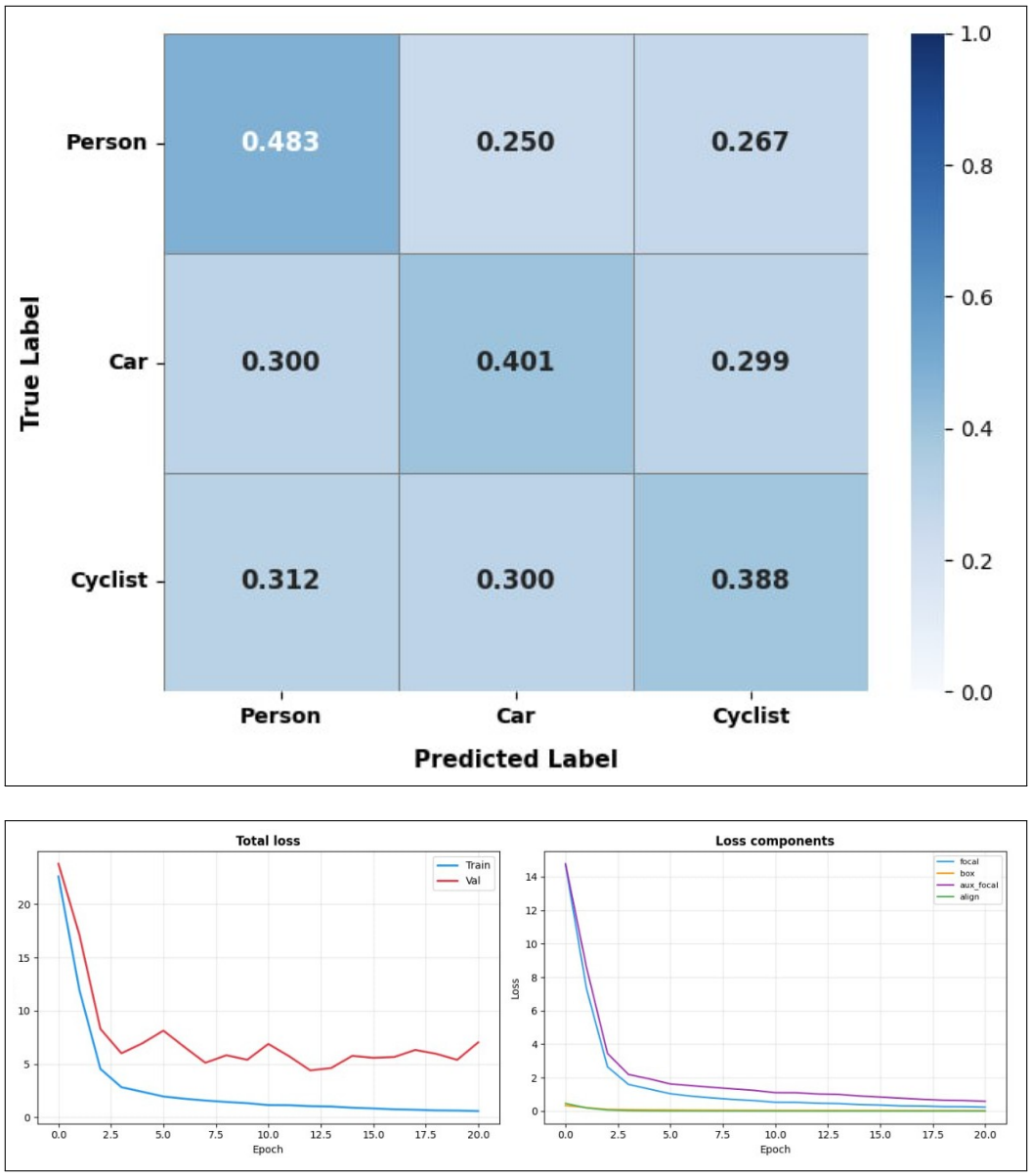


Figure 4.6: RAMP-CNN+ evaluation without Stage 1 super-resolution. The upper figure shows the row-normalized confusion matrix, and the lower figure shows the training and validation loss curves together with the loss components.

The per-class detection results of the RAMP-CNN+ detector without Stage 1 super-resolution are summarized in Table 4.2. In this configuration, the model achieves F1 scores of 0.497 for person, 0.426 for car, and 0.426 for cyclist.

Table 4.2: Per-class RAMP-CNN+ detection results on the test set without Stage 1 super-resolution.

Class	TP	FP	FN	Precision	Recall	F1
Person	4,034	3,845	4,318	0.512	0.483	0.497
Car	478	573	714	0.455	0.401	0.426
Cyclist	2,348	2,616	3,704	0.473	0.388	0.426

The aggregate results without Stage 1 super-resolution are given in Table 4.3. The detector achieves a weighted F1 score of 0.468, mAP of 0.480, and mAR of 0.424.

Table 4.3: Aggregate RAMP-CNN+ detection results on the test set without Stage 1 super-resolution.

Metric	Value
Weighted F1	0.468
mAP	0.480
mAR	0.424

The same detector was then evaluated using the SR-enhanced RA representation produced by Stage 1. The SR-enhanced configuration is reported in Table 4.4 and Table 4.5.

Table 4.4: Per-class RAMP-CNN+ detection results on the test set with Stage 1 super-resolution.

Class	TP	FP	FN	Precision	Recall	F1
Person	4,866	1,600	1,666	0.753	0.745	0.749
Car	580	250	280	0.699	0.674	0.686
Cyclist	2,746	1,100	1,638	0.714	0.626	0.667

The aggregate RAMP-CNN+ metrics are summarized in Table 4.5. The RAMP-CNN+ detector achieves a weighted F1 score of 0.717, mAP of 0.722, and mAR of 0.682. These results indicate that the model maintains strong recall while also achieving a more balanced precision–recall profile across the retained classes.

Table 4.5: Aggregate RAMP-CNN+ detection results on the test set with Stage 1 super-resolution.

Metric	Value
Weighted F1	0.717
mAP	0.722
mAR	0.682

The per-class recall values are 0.745 for person, 0.674 for car, and 0.626 for cyclist. The corresponding precision values are 0.753, 0.699, and 0.714, respectively. Thus, the model shows a reasonably stable detection profile across all three retained classes, although cyclist remains the most difficult class in terms of recall and F1 score.

Compared with the raw-RA configuration, the SR-enhanced configuration improves the weighted F1 score from 0.468 to 0.717, mAP from 0.480 to 0.722, and mAR from 0.424 to 0.682. This indicates that, under this experimental setting, the SR-enhanced RA input is associated with improved RAMP-CNN+ detection performance. However, the comparison is interpreted within the implemented pipeline and dataset, and the broader relationship between super-resolution and detection remains subject to the matched-geometry limitations discussed in Section 4.4.5.

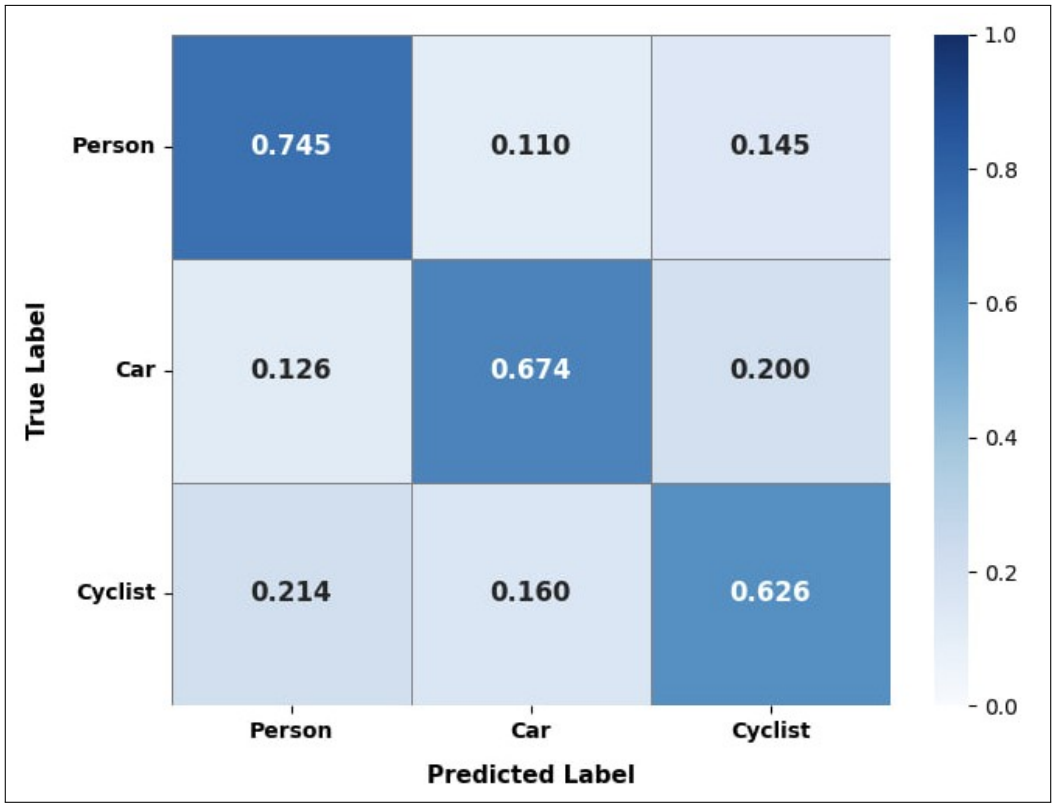


Figure 4.7: Confusion matrix for the RAMP-CNN+ detector with Stage 1 super-resolution.

4.3.4 Prototype-Based Classification and Counting Results

The prototype-based `FullModel` is evaluated separately from RAMP-CNN+ because it does not produce heatmap detections. Instead, it produces class logits of shape $[B, 3]$ and object-count predictions of shape $[B, 4]$. The four count bins correspond to zero, one, two, and three or more objects in the scene, as described in Section 3.5. In this experiment, the prototype-based classification model is compared in two configurations: without the Stage 1 super-resolution input and with the SR-enhanced RA input.

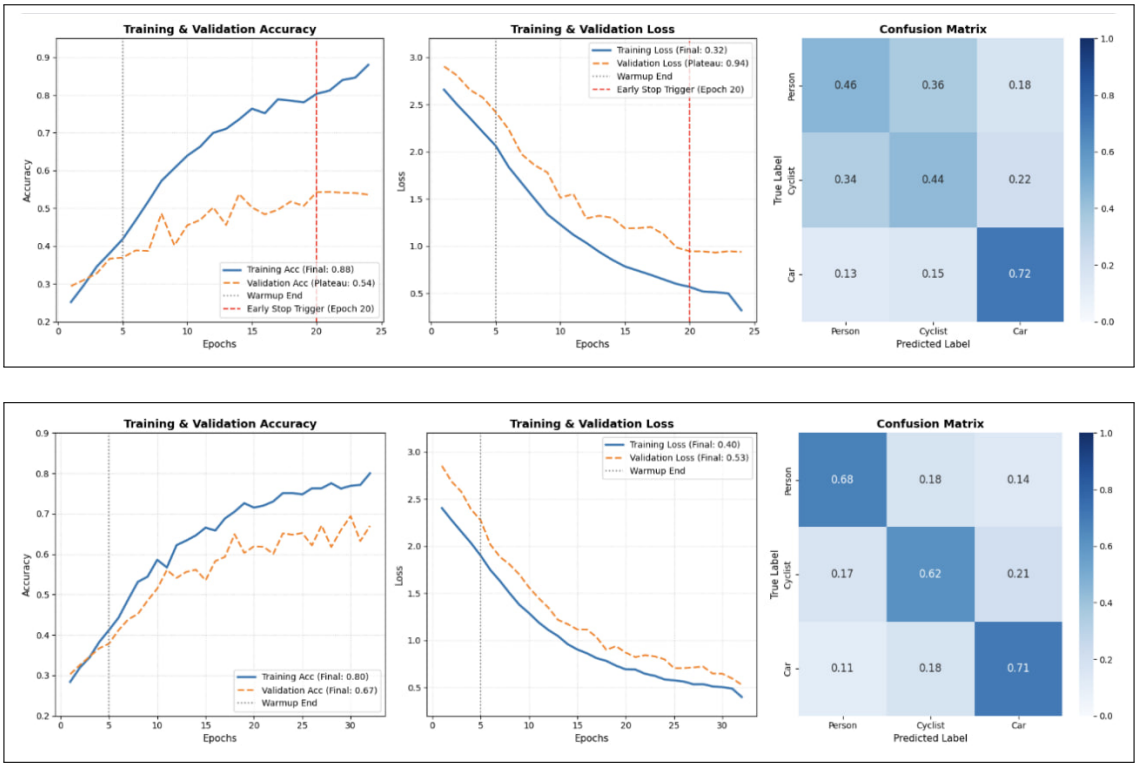


Figure 4.8: Training behaviour and row-normalized confusion matrices of the prototype-based `FullModel`. The upper figure corresponds to the configuration without Stage 1 super-resolution, while the lower figure corresponds to the configuration using the SR-enhanced RA representation.

The training behaviour and confusion matrices are summarized in Figure 4.8. Without the SR-enhanced RA input, the model reaches a final training accuracy of 0.88, but the validation accuracy plateaus at 0.54. The final training loss is 0.32, while the validation loss plateaus at 0.94. This indicates a clear gap between training and validation performance and suggests stronger overfitting in the non-SR configuration.

With the SR-enhanced RA input, the model reaches a final training accuracy of 0.80 and a validation accuracy of 0.67. The final training loss is 0.40, while the validation loss decreases to 0.53. Although the training accuracy is lower than in the non-SR configuration, the validation accuracy is higher and the validation loss is substantially lower. This indicates better

generalization when the prototype-based classifier uses the SR-enhanced RA representation.

Table 4.6: Prototype-based `FullModel` classification comparison with and without Stage 1 super-resolution.

Configuration	Train Acc.	Val. Acc.	Train Loss	Val. Loss
Without SR	0.88	0.54	0.32	0.94
With SR	0.80	0.67	0.40	0.53

The row-normalized confusion matrices further show that the SR-based configuration improves the correct classification rate for person and cyclist. For person, the diagonal value increases from 0.46 without SR to 0.68 with SR. For cyclist, it increases from 0.44 to 0.62. The car class remains nearly unchanged, with a diagonal value of 0.72 without SR and 0.71 with SR. This suggests that the main benefit of the SR-enhanced representation in the prototype-based classifier appears in the more difficult person and cyclist classes, while car classification remains relatively stable.

Table 4.7: Class-wise correct classification rates from the row-normalized confusion matrices of the prototype-based `FullModel`.

Class	Without SR	With SR	Change
Person	0.46	0.68	$+0.22$
Cyclist	0.44	0.62	$+0.18$
Car	0.72	0.71	-0.01

Overall, the prototype-based model shows improved validation behaviour when the SR-enhanced RA representation is used. The validation accuracy increases from 0.54 to 0.67, and the validation loss decreases from 0.94 to 0.53. Therefore, within this prototype-based classification experiment, the SR-enhanced RA input is associated with better generalization and stronger discrimination of person and cyclist classes. This comparison should be interpreted specifically for the scene-level prototype-based classifier, and not as a universal claim that super-resolution alone explains all Stage 2 detection behaviour.

4.3.5 Prediction Overlays

Qualitative prediction overlays are used to inspect the spatial behaviour of the RAMP-CNN+ detector. These visualizations compare the radar views, ground-truth heatmaps, predicted heatmaps, detections, and error maps for selected test frames. They are useful for identifying whether errors are caused by missed detections, duplicate detections, spurious edge responses, or spatially shifted predictions.

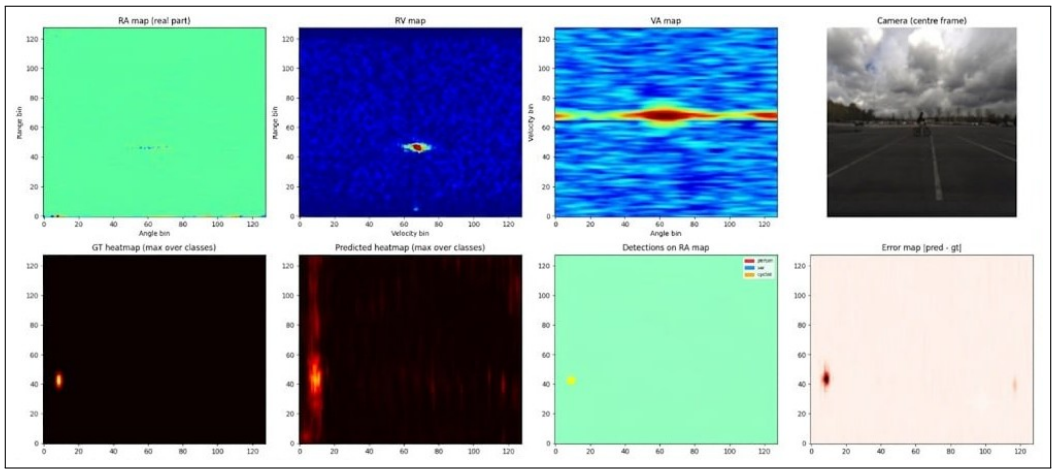


Figure 4.9: Example RAMP-CNN+ prediction overlay for a held-out test frame.

4.4 Discussion

The Stage 1 and Stage 2 results demonstrate two distinct parts of the proposed radar perception pipeline. Stage 1 shows that a learned super-resolution model can reconstruct a dense-array-equivalent range–azimuth representation with high image fidelity on held-out data and can qualitatively sharpen measured real-world radar frames. Stage 2 shows that multi-view radar inputs can be used for three-class road-user detection and classification with a balanced precision–recall profile.

4.4.1 Interpretation of Stage 1 Results

The Stage 1 quantitative results indicate strong reconstruction performance. The validation PSNR of 42.27 dB and test PSNR of 40.10 dB show low reconstruction error relative to the dense-array target. The validation SSIM of 0.9463 and test SSIM of 0.8716 further indicate that the spatial structure of the dense-array RA map is preserved. These results support the use of the Attention U-Net as an angular super-resolution front-end.

The qualitative real-world example is also important because the model is trained using simulation and synthetic data. In the measured radar example, the raw sparse-array RA map is dominated by angular aliasing and side-lobe artifacts, while the SR output produces cleaner focused peaks aligned with the camera scene. This supports the practical usefulness of Stage 1 as an angular enhancement module. However, quantitative real-world PSNR/SSIM cannot be computed without measured dense-array ground truth.

4.4.2 Precision–Recall Balance in RAMP-CNN+

The RAMP-CNN+ detector achieves a balanced precision–recall profile on the test set. The aggregate mAR is 0.682, while the mAP is 0.722 and the weighted F1 score is 0.717. Compared with the raw-RA configuration, the SR-enhanced configuration improves weighted F1 from 0.468 to 0.717, mAP from 0.480 to 0.722, and mAR from 0.424 to 0.682 under the implemented evaluation setting.

The per-class recall values are 0.745 for person, 0.674 for car, and 0.626 for cyclist. The corresponding precision values are 0.753, 0.699, and 0.714, respectively. Thus, the model shows a reasonably stable detection profile across all three retained classes. Cyclist remains the most challenging class because it has the lowest recall and F1 score among the retained classes.

Although the final precision values are substantially improved, false-positive control remains important for practical deployment. Further threshold tuning, stronger non-maximum suppression, per-class thresholds, or edge-aware suppression may still improve the detector's reliability in more complex driving scenes.

4.4.3 Class-Wise Behaviour

Among the three classes, person achieves the strongest F1 score of 0.749, with precision 0.753 and recall 0.745. Car achieves an F1 score of 0.686, with precision 0.699 and recall 0.674. Cyclist achieves an F1 score of 0.667, with precision 0.714 and recall 0.626.

The results show that all three classes achieve comparable detection performance, with person performing best overall. Cyclist remains the most challenging class because it has the lowest recall and F1 score among the retained classes.

4.4.4 Scene Dependence and Radar-Only Detection Difficulty

The evaluated Stage 2 scenes are largely open parking-lot and road-scene environments. Such environments can produce weak, diffuse, or ambiguous radar returns. Radar data also has lower semantic richness than camera images and lower spatial density than lidar point clouds. Therefore, absolute mAP values for radar-based three-class detection should be interpreted carefully, especially when evaluated on a specific dataset and operating point.

The mAP of 0.722 and mAR of 0.682 indicate that the detector achieves a reasonable balance between precision and recall on the held-out test set. However, these values should still be interpreted in the context of the specific dataset, sensor setup, class definition, and matching rule used in this thesis. Broader evaluation on more diverse driving scenes would be required before making a general deployment claim.

A direct numerical comparison with other radar detection studies is not made in this chapter unless the datasets, sensor setups, label definitions, and evaluation criteria are matched. Radar perception performance can vary significantly with scene structure, sensor configuration, annotation policy, and matching rule. For this reason, the results are interpreted primarily as a controlled evaluation of the implemented pipeline rather than as a universal benchmark claim.

4.4.5 Relationship between Super-Resolution and Detection

The results of the two stages should be interpreted separately and honestly. Stage 1 provides evidence that the super-resolution network improves the visual and structural quality of the RA representation. Stage 2 provides evidence that the multi-view RAMP-CNN+ detector can detect and classify three road-user classes with a balanced precision–recall profile.

This thesis does not claim that the measured Stage 2 metric improvements are caused solely by the super-resolution module. The Stage 2 performance is also strongly affected by the model architecture, loss formulation, training stabilization, multi-view fusion, and evaluation threshold. A fully controlled study isolating the contribution of super-resolution to downstream detection would require identical detectors trained and evaluated on raw and super-resolved RA inputs under matched array geometry. That study is left as future work.

4.4.6 Summary of Results

Overall, the experimental results support the feasibility of the proposed two-stage radar perception pipeline. Stage 1 achieves high-fidelity angular super-resolution, with test PSNR of 40.10 dB and test SSIM of 0.8716, and demonstrates qualitative sim-to-real enhancement on measured radar data. Stage 2 achieves weighted F1 of 0.717, mAP of 0.722, and mAR of 0.682 on the held-out test set using the heatmap-based RAMP-CNN+ detector with the SR-enhanced RA input. Compared with the raw-RA configuration, where weighted F1, mAP, and mAR are 0.468, 0.480, and 0.424, respectively, the SR-enhanced configuration shows improved detection performance under the implemented experimental setting. The detector shows a balanced precision–recall profile across the retained person, car, and cyclist classes, with cyclist remaining the most challenging class in terms of recall and F1 score. The prototype-based model provides an additional classification and object-counting pathway. In the prototype-based classification experiment, the SR-enhanced configuration improves validation accuracy from 0.54 to 0.67 and reduces validation loss from 0.94 to 0.53, indicating better generalization than the non-SR configuration. These results identify both the promise of the proposed pipeline and the main directions for future improvement.

CHAPTER 5: EVALUATION, LIMITATIONS, FUTURE WORK, AND CONCLUSION

This chapter summarizes the overall evaluation of the proposed two-stage radar perception pipeline. The chapter first discusses the major limitations observed in the implemented system, including Stage 2 false positives, overfitting risk, scene-dependence, and super-resolution array-geometry compatibility. It then outlines possible future extensions and concludes the thesis by restating the main technical outcomes of the two stages.

5.1 Limitations

Although the proposed pipeline demonstrates promising results for radar super-resolution and radar-based road-user perception, several limitations remain. These limitations are important for interpreting the results in Chapter 4 and for defining the scope of the conclusions drawn from this thesis.

5.1.1 Stage 2 False Positives and Deployment Robustness

The heatmap-based Stage 2 detector achieves a balanced precision–recall profile on the held-out test set when the SR-enhanced RA representation is used, with weighted F1 of 0.717, mAP of 0.722, and mAR of 0.682. The final per-class precision values are 0.753 for person, 0.699 for car, and 0.714 for cyclist. The corresponding recall values are 0.745, 0.674, and 0.626, respectively. These results show that the detector is able to identify a substantial portion of true objects while maintaining reasonable precision across the retained classes. Compared with the raw-RA configuration, where weighted F1, mAP, and mAR are 0.468, 0.480, and 0.424, respectively, the SR-enhanced configuration shows improved detection performance under the implemented evaluation setting.

However, false-positive control remains an important limitation for practical deployment. Even when aggregate precision improves, radar detections can still be affected by clutter, ambiguous reflections, edge responses, or duplicate peaks. In an autonomous-driving system, repeated false detections may affect downstream tracking, planning, or decision-making modules. Therefore, additional post-processing and deployment-level validation are still required before the detector can be considered robust for real-world use.

The current operating point provides a reasonable balance between precision and recall, but further tuning may still be necessary for different applications. A safety-oriented system may prefer higher recall, while a tracking or planning system may require stricter false-positive

suppression. Therefore, per-class thresholds, improved non-maximum suppression, and edge-aware filtering remain useful future improvements.

5.1.2 Overfitting and Limited Training Windows

The Stage 2 detector contains a large number of trainable parameters and is trained using overlapping temporal radar windows. Because adjacent frames in a driving sequence are highly correlated, the number of effective independent training examples is smaller than the raw number of windows may suggest. This creates a risk of overfitting, especially for a model with a high-capacity multi-view architecture.

The final RAMP-CNN+ training run showed mild overfitting after the best validation epoch: the training loss continued to decrease while the validation loss plateaued or increased. Early stopping was used to retain the best checkpoint and reduce the effect of this overfitting. Nevertheless, the result indicates that stronger regularization and broader training data would be useful in future work.

5.1.3 Open-Scene and Dataset Dependence

The Stage 2 dataset consists mainly of open parking-lot and road-scene environments. These scenes are useful for controlled radar perception experiments, but they do not fully represent the complexity of dense urban driving. Urban scenes may contain stronger multipath, more occlusion, more static clutter, larger numbers of road users, and more diverse object arrangements.

As a result, the reported Stage 2 performance should be interpreted as a dataset-specific evaluation rather than a general claim about all automotive radar scenes. The model is restricted to the retained three-class setup: person, car, and cyclist. Low-frequency or unused categories were excluded to obtain a stable three-class training and evaluation problem. Therefore, the model should not be interpreted as a complete all-object automotive perception system.

The prototype-based classification model also has a limited scope. It is designed for scene-level classification and object counting rather than spatial object localization. Its results should therefore be interpreted separately from the heatmap-based RAMP-CNN+ detector.

5.1.4 Super-Resolution Array-Geometry Compatibility

The Stage 1 super-resolution model is trained for a specific sparse-to-dense virtual-array mapping, namely from a $2\mathrm{Tx} \times 4\mathrm{Rx}$ sparse configuration to a $5\mathrm{Tx} \times 10\mathrm{Rx}$ dense configuration. This geometry-specific training is appropriate for the supervised super-resolution formulation, but it also limits direct generalization to other antenna layouts. A model trained for one virtual-array geometry may not directly transfer to another geometry without retraining or adaptation.

The super-resolution network is trained for a specific virtual-array geometry. In this thesis, the implemented RAMP-CNN+ detector is evaluated using both raw RA input and SR-enhanced RA input under the same evaluation protocol, and the SR-enhanced configuration shows improved detection metrics. However, this comparison should still be interpreted within the implemented dataset, model, and array-geometry setting. A broader controlled study across additional scenes, radar geometries, and repeated training runs would be needed before making a general claim that super-resolution alone determines downstream detection performance.

This limitation is important for honest interpretation of the two-stage pipeline. Stage 1 demonstrates angular enhancement using PSNR, SSIM, and qualitative real-world comparison. Stage 2 demonstrates radar-based detection and classification using the implemented multi-view architectures. However, the current results should not be overstated as proof that super-resolution alone causes the measured Stage 2 detection performance.

5.1.5 Simulation-Based Super-Resolution Supervision

The Stage 1 network is trained using Dataset3000 and additional synthetic point-target data. Dataset3000 provides paired sparse- and dense-array ADC data by rendering the same CARLA target list through two radar configurations. This is a practical and defensible method for obtaining supervised sparse/dense pairs, since perfectly matched measured dense-array ground truth is difficult to collect.

However, simulation-based radar data cannot perfectly reproduce all real-world effects. Real radar measurements may include sensor imperfections, calibration errors, multipath, material-dependent scattering, environmental clutter, and near-field effects that are not fully represented in the analytic signal model. The qualitative measured-data result shows that the Stage 1 model can produce cleaner angular peaks on real radar frames, but quantitative PSNR/SSIM on real measured dense-array targets cannot be reported without measured dense-array ground truth.

5.2 Future Work

The limitations discussed above suggest several directions for future work.

5.2.1 Matched-Geometry Super-Resolution and Controlled A-vs-B Detection

A major future direction is to extend the raw-RA versus SR-enhanced-RA comparison under a broader matched array-geometry setting. The present thesis compares the implemented RAMP-CNN+ detector with and without the Stage 1 SR-enhanced RA input, while keeping the evaluation protocol consistent. Future work should repeat this comparison across additional radar

scenes, random seeds, array configurations, and model variants to determine whether the observed improvement remains stable under broader conditions.

This extended experiment would help identify whether super-resolution improves not only the visual quality of the RA map but also downstream detection metrics across a wider range of operating conditions. It would strengthen the end-to-end claim of the two-stage pipeline while preserving a clear distinction between image-level SR improvement and detection-level performance.

5.2.2 False-Positive Reduction

One important future improvement for RAMP-CNN+ is further false-positive reduction and deployment-level robustness testing. Several inference-time and training-time techniques may be investigated. Edge masking could suppress detections near problematic angle-axis boundaries. Stronger non-maximum suppression could reduce duplicate or clustered detections. Per-class thresholds could allow stricter decision boundaries for classes or scenes that produce repeated false detections. An edge-aware loss term could also be explored to discourage repeated false detections in known difficult regions.

These methods should be evaluated carefully because reducing false positives can also reduce recall. The desired precision–recall balance depends on the intended use of the system. For safety-critical perception, high recall may remain preferable, but precision must still be improved enough to avoid excessive false alarms.

5.2.3 Full Ablation of RAMP-CNN+ Contributions

The RAMP-CNN+ model includes four main contributions: complex-valued RA processing, cross-modal feature alignment, squeeze-and-excitation adaptive fusion, and class-balanced asymmetric focal loss. A full ablation study should train reduced model variants to quantify the contribution of each component. Possible variants include a baseline model, a model with class-balanced loss only, a model with adaptive fusion, a model with camera alignment, and the full model.

Such an ablation would make the contribution of each architectural and training component more explicit. It would also help identify which parts of the model are most important for recall, precision, and class-wise performance.

5.2.4 Broader Dataset Evaluation

Future work should evaluate the proposed models on broader and more diverse radar datasets. Additional scenes should include dense urban roads, intersections, cluttered roadside environ-

ments, adverse weather, night-time conditions, and more varied traffic densities. Broader evaluation would help determine whether the model generalizes beyond the open-scene dataset used in this thesis.

The class set should also be expanded when sufficient labelled examples are available. Including additional road users and vehicle types would make the perception task closer to real automotive deployment. However, such expansion should be done with sufficient class balance to avoid unstable training and misleading evaluation.

5.2.5 Real-Time and Embedded Implementation

The present work is implemented as an offline research pipeline using cached radar views and Kaggle GPU notebooks. Future work should measure end-to-end inference time on the target deployment hardware. The pipeline may need model compression, pruning, quantization, or lightweight backbone redesign before real-time embedded deployment.

A streaming implementation should process incoming ADC frames continuously, generate RA, RV, and VA representations online, apply the Stage 1 super-resolution module, and run the Stage 2 detector or classifier within a fixed latency budget. Such real-time evaluation is necessary before the system can be considered for practical autonomous-driving use.

5.3 Conclusion

This thesis presented a two-stage radar perception pipeline for autonomous-driving applications. The first stage addressed the cost–resolution tradeoff in automotive radar by learning angular super-resolution from sparse-array range–azimuth data. The second stage used multi-view radar representations for road-user detection, classification, and object-counting tasks.

Stage 1 used an Attention U-Net to map sparse $2\mathrm{Tx} \times 4\mathrm{Rx}$ radar features to a dense-array-equivalent $5\mathrm{Tx} \times 10\mathrm{Rx}$ range–azimuth representation. The model used five-channel radar features containing beamsteering and spatial-covariance information. On held-out test data, the Stage 1 model achieved a PSNR of 40.10 dB and an SSIM of 0.8716. The best validation result reached a PSNR of 42.27 dB and an SSIM of 0.9463. Qualitative evaluation on independently measured real-world radar showed that the SR output produced cleaner and more focused angular peaks than the raw sparse-array RA map.

Stage 2 evaluated radar-based perception using multi-view RA, RV, and VA representations. The heatmap-based RAMP-CNN+ detector combined complex-valued RA processing, cross-modal feature alignment, adaptive multi-view fusion, and a class-balanced asymmetric focal loss. With the SR-enhanced RA input, RAMP-CNN+ achieved a weighted F1 score of 0.717, mAP of 0.722, and mAR of 0.682 on the held-out test set. In comparison, the raw-RA con-

figuration achieved weighted F1 of 0.468, mAP of 0.480, and mAR of 0.424. Thus, within the implemented experimental setting, the SR-enhanced configuration showed better detection performance than the raw-RA configuration. The per-class F1 scores for the SR-enhanced configuration were 0.749 for person, 0.686 for car, and 0.667 for cyclist, with cyclist remaining the most challenging class in terms of recall and F1 score. Further false-positive reduction and broader dataset evaluation remain important future directions.

A second Stage 2 model, `FullModel`, was also formulated for prototype-based classification and object counting. This model used multi-scale scene features, projection-based radar encodings, cross-projection attention, and EMA class prototypes. It provides a complementary classification and counting pathway beside the heatmap-based detector.

Overall, the thesis demonstrates that deep learning can be used to enhance sparse-array radar representations and to perform multi-view radar perception for selected road-user classes. The work also identifies important engineering constraints, including geometry-specific super-resolution training, limited dataset diversity, false positives, and real-time deployment requirements. These findings provide a foundation for future work on matched-geometry SR integration, broader raw-versus-SR detection experiments, broader dataset evaluation, and real-time radar perception systems.

REFERENCES

[1] Patole, S.M., Torlak, M., Wang, D. and Ali, M. "Automotive radars: A review of signal processing techniques." *IEEE Signal Processing Magazine*, volume 34, no. 2:pp. 22–35, 2017

[2] Liu, Z., Cai, Y., Wang, H., Chen, L., Gao, H., Jia, Y. and Li, Y. "Robust target recognition and tracking of self-driving cars with radar and camera information fusion under severe weather conditions." *IEEE Transactions on Intelligent Transportation Systems*, volume 23, no. 7:pp. 6640–6653, 2021

[3] Zhang, X., Xu, L., Xu, L. and Xu, D. "Direction of departure (DOD) and direction of arrival (DOA) estimation in mimo radar with reduced-dimension music." *IEEE Communications Letters*, volume 14, no. 12:pp. 1161–1163, 2010

[4] Oh, D., Li, Y.c., Khodjaev, J., Chong, J.W. and Lee, J.H. "Joint estimation of direction of departure and direction of arrival for multiple-input multiple-output radar based on improved joint esprit method." *IET Radar, Sonar & Navigation*, volume 9, no. 3:pp. 308–317, 2015

[5] Potter, L.C., Ertin, E., Parker, J.T. and Cetin, M. "Sparsity and compressed sensing in radar imaging." *Proceedings of the IEEE*, volume 98, no. 6:pp. 1006–1020, 2010

[6] Correas-Serrano, A. and González-Huici, M.A. "Experimental evaluation of compressive sensing for doa estimation in automotive radar." In "2018 19th International Radar Symposium (IRS)," pp. 1–10. IEEE, 2018

[7] Schuessler, C., Hoffmann, M. and Vossiek, M. "Super-resolution radar imaging with sparse arrays using a deep neural network trained with enhanced virtual data." *IEEE Journal of Microwaves*, volume 3, no. 3:pp. 980–993, 2023. ArXiv:2306.09839

[8] Gao, X., Xing, G., Roy, S. and Liu, H. "Ramp-cnn: A novel neural network for enhanced automotive radar object recognition." *IEEE Sensors Journal*, volume 21, no. 4:pp. 5119–5132, 2021

[9] Wang, Y., Jiang, Z., Li, Y., Hwang, J.N., Xing, G. and Liu, H. "Rodnet: A real-time radar object detection network cross-supervised by camera-radar fused object 3d localization." *IEEE Journal of Selected Topics in Signal Processing*, volume 15, no. 4:pp. 954–967, 2021

[10] Wu, Z., Chen, G., Gan, Y., Wang, L. and Pu, J. "Mvfusion: Multi-view 3d object detection with semantic-aligned radar and camera fusion." *arXiv preprint arXiv:2302.10511*, 2023

[11] Du, R., Fan, Y. and Wang, J. “Pedestrian and bicyclist identification through micro doppler signature with different approaching aspect angles.” *IEEE Sensors Journal*, volume 18, no. 9:pp. 3827–3835, 2018

[12] Molchanov, P., Astola, J., Egiazarian, K. and Totsky, A. “Classification of ground moving radar targets by using joint time-frequency analysis.” In “2012 IEEE Radar Conference,” pp. 0366–0371. IEEE, 2012

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Deep Sparse-to-Dense Automotive Radar Perception for Super-Resolved Object Detection and Classification

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June

2026